

Weighted Odd-Step Paths, Survivor Coding, Arithmetic Groupoids, and Toric Period Morphisms

A Research Companion for the Collatz Map

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proof reconstruction and literature crosswalk

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Abstract

We retain the positive-odd first-return coordinates of the Collatz map and construct their legal block category, coefficientwise path completion, predecessor transfer operator, injective decorated prefix embedding, and finite residue transformation semigroups. Exact comparison maps separate two $\mathbb{F}_2^2$ gradings, the prefix coordinate from the forward affine intercept, and forward actions from contravariant pullback. The finite semigroups receive constructive normal forms, cardinalities, ranks, minimum ideals, reduction fibres, and a coefficient-level 3-adic inverse limit.

The same prefix pieces define a Gaussian relation lattice for rational arctangent angles. We construct its exact signed-valuation morphism, prove an all-word non-torsion theorem for the unweighted sum, determine the exceptional two-letter lattice, and give a coefficient-unbounded exact certificate for every prefix of length at most fifteen and cumulative exponent at most sixty. The resulting all-length rigidity statement is isolated as a conjecture and crosswalked against the published Gaussian and Machin literature.

The finite prefix classes then give a homeomorphism from the full countable label shift to the odd 2-adic survivor space obtained by deleting the backward orbit of $-1/3$. We locate the positive-integer image, prove its canonical-representative stabilization criterion, and compute the exact full-shift pressure. Finally, object-idempotent and formal-scheme arguments give precise boundaries for proposed tropical and prismatic interfaces.

The same path coordinates define a discrete arithmetic orbit groupoid with two exact cocycles and a faithful path-algebra embedding. Exact homeomorphisms of $\mathbb{Z}_3$ generate a compact-unit germ groupoid, and a proved affine-germ functor carries each arithmetic orbit arrow into it while retaining its bidegree and real weight. A unimodular torus map then turns the bidegree into the arithmetic character, yielding commutative 1-motive squares, Kummer periods, a Cayley isomorphism of algebraic pairs, and the contravariant Nori edges induced by actual pair maps.

Finally, positive exponent compositions are identified exactly with pointed shortened-map parity words. The resulting groupoid isomorphism carries packet rotation to rotation by the corresponding exponent prefix, and the classical rational-cycle numerator becomes the odd-step affine numerator without a change of convention. The primitive packet of length d and exponent sum B has exact periods d , B , and $B + d$ under the odd-return, shortened, and full maps, respectively. A finite-state residue recursion and two independent integer-only scans certify all strip-selected packets through odd-step period sixteen, including the complete 99-class period-eight table. The Chinese remainder coordinate map intertwines the residue transitions letter by letter. Its synchronized product retains the common exponent word, whereas marginalization has an explicit rectangle-basis kernel. The complete length-ten,

sum-sixteen residue table separates primitive words from repeated words and records their different divisor exclusions.

The packet affine offsets also admit a closed two-state generating family. The corresponding fixed-point series is uniquely characterized coefficientwise by a dyadic dilation equation. We determine its exact Taylor domains, construct a normally convergent meromorphic continuation, and compute every finite pole, its order, and its leading coefficient.

Every nonconjectural Collatz-specific assertion is proved in the paper; portable finite certificates independently check the bounded algebra, residue, symbolic, pressure, Gaussian-relation, groupoid, affine-germ, torus-character, and pair-map calculations. The results do not assert termination of all positive Collatz orbits.

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1 Weighted odd-step paths, exact completions, and finite residue actions

This separately authored research companion develops a theorem package suggested by a chronological collection of Collatz research notes. The notes are intellectual provenance, not theorem authority: every statement retained here is rederived from the displayed definitions. The starting sketches contained a substantial finite core, together with four precise defects: mutually incompatible composition orders, nonunital transformation semigroups called monoids, infinite path sums placed in a finite-support algebra, and a prefix 3-adic cocycle identified with a forward affine intercept without reversing the exponent word. The constructions below keep the original odd-step coordinates and repair those defects at the point where they occur.

The standard algebraic vocabulary is crosswalked against the definitions of free monoids, semirings, formal power series, weighted paths, and transformation monoids in [9, 8, 10]. Those sources do not contain the Collatz-specific theorems below; every such theorem is proved here. No claim of priority is made for results independently reconstructed in this paper.

1.1 The odd first-return category and its local block category

Let

$$\mathcal{O} = \{1, 3, 5, \dots\}, \quad a(x) = \nu_2(3x + 1), \quad C(x) = \frac{3x + 1}{2^{a(x)}} \quad (x \in \mathcal{O}).$$

Thus C is the positive-odd first-return map. Write

$$o(n) = \frac{3n + 1}{2} \quad (n \text{ odd}), \quad e(n) = \frac{n}{2} \quad (n \text{ even})$$

for the two legal branches of the shortened map, with their domains retained.

Definition 1.1 (first-return path category). The category $\mathcal{E}_C$ has object set $\mathcal{O}$. For every $x \in \mathcal{O}$ it has a generating arrow

$$\epsilon_x : x \longrightarrow C(x), \quad \lambda(\epsilon_x) = a(x),$$

and its morphisms are finite composable strings of these arrows, including the empty path 1_x at each object. A path

$$p : x_0 \xrightarrow{a_1} x_1 \xrightarrow{a_2} \dots \xrightarrow{a_n} x_n$$

has length and total exponent

$$\ell(p) = n, \quad A(p) = a_1 + \dots + a_n,$$

and weight

$$w(p) = u^{\ell(p)} v^{A(p) - \ell(p)}.$$

The category $\mathcal{D}_C^{\text{blk}}$ has the same objects. Its morphisms are precisely the legal shortened-map paths between odd endpoints whose local word has the form

$$(oe^{m_1})(oe^{m_2}) \cdots (oe^{m_n}), \quad m_i \geq 0.$$

For a block path q , put

$$\#_o(q) = \text{number of } o\text{-steps}, \quad \#_e(q) = \text{number of } e\text{-steps}, \quad w_{\text{blk}}(q) = u^{\#_o(q)} v^{\#_e(q)}.$$

Composition in both categories is path concatenation.

The set of strict first-return edges by itself is a directed graph, not a category: a composite of two strict first-return edges is usually not a single strict first-return edge. The free path category $\mathcal{E}_C$ is the typed categorical object used below.

Theorem 1.2 (exact odd-block category isomorphism). *For a positive exponent put*

$$\iota(a) = oe^{a-1}.$$

There is a functor

$$\iota : \mathcal{E}_C \longrightarrow \mathcal{D}_C^{\text{blk}}$$

which is the identity on objects and replaces each exponent edge by its local block. It is an isomorphism. Its inverse is

$$U((oe^{m_1}) \cdots (oe^{m_n})) = (m_1 + 1, \dots, m_n + 1).$$

Moreover

$$\#_o(\iota(p)) = \ell(p), \quad \#_e(\iota(p)) = A(p) - \ell(p), \quad w_{\text{blk}}(\iota(p)) = w(p).$$

In particular, this block passage loses no local trajectory information.

Proof. Fix $x \in \mathcal{O}$ and put $a = a(x)$. The first local step gives $(3x + 1)/2$. If $a > 1$, this value and the following $a - 2$ intermediate values are even, so exactly $a - 1$ legal e -steps follow. Their endpoint is $(3x + 1)/2^a = C(x)$, which is odd. Conversely, if a legal local path from an odd x has word oe^m and first returns to an odd state at its endpoint, then $3x + 1$ is divisible by 2^{m+1} and not by 2^{m+2} . Hence $a(x) = m + 1$.

Every displayed block word has a unique parsing: each occurrence of o starts a block and every following run of e 's ends immediately before the next o or at the word endpoint. Therefore $U\iota$ and ιU are the identity on every morphism. Both maps preserve endpoints and concatenation, so they are mutually inverse functors. Counting the one o and the $a_i - 1$ copies of e in each block proves the three weight identities. $\square$

The theorem corrects a recurrent description of U as a proper forgetful quotient. It would be information-losing on an unrestricted local word category, but on the explicitly defined legal odd-to-odd block category it has the displayed two-sided inverse.

1.2 Two different $\mathbb{F}_2^2$ degrees

The weighted block counts give a degree

$$d_{\text{count}}(p) = (\ell(p) \bmod 2, (A(p) - \ell(p)) \bmod 2).$$

A separate coefficient-label convention assigns 00 to every even branch and 11 to every odd branch. On an odd-to-odd block path it instead gives

$$d_{\text{coeff}}(p) = (\ell(p) \bmod 2, \ell(p) \bmod 2).$$

Proposition 1.3 (exact comparison of the two degrees). *The homomorphism*

$$\kappa : \mathbb{F}_2^2 \longrightarrow \mathbb{F}_2^2, \quad \kappa(r, s) = (r, r),$$

satisfies $d_{\text{coeff}} = \kappa \circ d_{\text{count}}$. Its kernel, image, and fibres are

$$\begin{aligned} \ker \kappa &= \{00, 01\}, & \text{im } \kappa &= \{00, 11\}, \\ \kappa^{-1}(00) &= \{00, 01\}, & \kappa^{-1}(11) &= \{10, 11\}, \end{aligned}$$

with empty fibres over 01 and 10. Thus κ is an idempotent projection onto the diagonal $\Delta = \{00, 11\}$. Equivalently, its codomain restriction induces the quotient isomorphism $\mathbb{F}_2^2 / \ker \kappa \simeq \Delta$. It is not a quotient onto the whole displayed codomain, and the two degrees are not interchangeable.

Proof. Substitution gives $\kappa(\ell, A - \ell) = (\ell, \ell)$ modulo two. The kernel, image, and four fibres follow by evaluating the four elements of $\mathbb{F}_2^2$. $\square$

1.3 Composition order and the ordinary category algebra

We use the standard target-left convention throughout this section. If $p : x \rightarrow y$ and $q : y \rightarrow z$, then $q \circ p$ means first p , then q , and the category-algebra product is

$$[q][p] = [q \circ p], \quad [q][p] = 0 \quad \text{when } t(p) \neq s(q). \quad (1)$$

Consequently

$$[p] = e_{t(p)}[p]e_{s(p)}, \quad e_x = [1_x],$$

and for the path in Definition 1.1,

$$[p] = [\epsilon_{x_{n-1}}] \cdots [\epsilon_{x_0}].$$

Chronological word concatenation is the opposite written order from this target-left product. Several raw formulas use one order in the path algebra and the other order in a transformation or pullback algebra without an opposite construction; Subsection 1.10 gives the forced repair.

Let R be a commutative ring. The ordinary weighted category algebra is

$$\mathfrak{A}_C = \bigoplus_{p \in \text{Mor}(\mathcal{E}_C)} R[u, v][p]$$

with multiplication (1). Every element has finite path support. Therefore neither the sum of all object idempotents nor the sum of all finite paths is generally an element of $\mathfrak{A}_C$.

1.4 Endpoint series and the exact path completion

For $x, y \in \mathcal{O}$, define

$$Z_{x \rightarrow y}(u, v) = \sum_{\substack{n \geq 0 \\ C^n(x) = y}} u^n v^{A_n(x) - n} \in R[v][[u]], \quad (2)$$

where $A_0(x) = 0$ and $A_n(x) = \sum_{j=0}^{n-1} a(C^j(x))$. The coefficient of each u^n is either zero or one monomial in v ; no convergence assertion is needed for this formal series.

Proposition 1.4 (exact hit-time alternatives). *For fixed $x, y \in \mathcal{O}$, the set*

$$H_{x,y} = \{n \geq 0 : C^n(x) = y\}$$

is empty, a singleton, or

$$r + d\mathbb{N}_0$$

for a unique first hit r and the least positive period d of y . In the last case, if

$$B = \sum_{j=0}^{d-1} a(C^j(y)),$$

then

$$Z_{x \rightarrow y}(u, v) = \frac{u^r v^{A_r(x) - r}}{1 - u^d v^{B - d}} \quad (3)$$

as a formal geometric series. Every positive d -cycle satisfies

$$2^B > 3^d. \quad (4)$$

Hence the numerical specialization $(u, v) = (3/2, 1/2)$ converges on every cyclic endpoint series. In contrast,

$$Z_{1 \rightarrow 1}(1, 1) = \sum_{n \geq 0} 1$$

diverges.

Proof. If $m < n$ are two hit times, then $C^{n-m}(y) = y$, so y is periodic. Once the orbit first reaches a periodic y , its later returns occur exactly at multiples of the least period. This proves the alternatives and the uniqueness of r, d . Each traversal of the cycle adds d to the length and B to the exponent sum, which gives (3).

Write the cycle as $y_i \xrightarrow{a_i} y_{i+1}$ with indices modulo d . For every i ,

$$2^{a_i} y_{i+1} = 3y_i + 1 > 3y_i.$$

Multiplication and cancellation of the positive product $\prod_i y_i$ give $2^B > 3^d$. At $(3/2, 1/2)$ the geometric ratio is $3^d/2^B < 1$. Finally $C(1) = 1$ and $a(1) = 2$, so every term of $Z_{1 \rightarrow 1}$ has weight $(uv)^n$. $\square$

The completion needed for global path sums is not the preceding direct sum.

Definition 1.5 (coefficientwise path completion). Put

$$\hat{\mathfrak{A}}_C = \prod_{p \in \text{Mor}(\mathcal{E}_C)} R[u, v][p].$$

For $F = (F_p)$ and $G = (G_p)$, define the coefficient at a path r by

$$(FG)_r = \sum_{r=q \circ p} F_q G_p.$$

This is a finite sum: a path of length n has exactly $n + 1$ cuts.

The finite-cut statement proves that multiplication is well defined for arbitrary path support. Associativity follows by identifying both parenthesizations with the finite set of two-cut decompositions of a fixed path. The completion contains

$$E = \sum_{x \in \mathcal{O}} e_x, \quad W = \sum_{x \in \mathcal{O}} uv^{a(x)-1}[\epsilon_x], \quad G = \sum_p w(p)[p].$$

Every nonempty path has a unique first edge and a unique last edge, so

$$G = E + WG = E + GW. \tag{5}$$

Because R is a ring, this says $G = (E - W)^{-1}$ inside the displayed completion. If S is instead a commutative semiring, repeat Definition 1.5 with coefficients $S[u, v]$. The finite-cut convolution and the identities $G = E + WG = E + GW$ remain valid there and define the Kleene star W^* ; the subtraction and inverse notation are not used. None of E, W, G is silently placed in an ordinary finite-support algebra.

1.5 The predecessor transfer operator

For odd y define

$$\mathcal{P}(y) = \left\{ a \geq 1 : x_a(y) = \frac{2^a y - 1}{3} \text{ is an integer} \right\}.$$

Lemma 1.6 (exact predecessor fibres). *For odd y ,*

$$\mathcal{P}(y) = \begin{cases} \emptyset, & y \equiv 0 \pmod{3}, \\ \{2, 4, 6, \dots\}, & y \equiv 1 \pmod{3}, \\ \{1, 3, 5, \dots\}, & y \equiv 2 \pmod{3}. \end{cases}$$

For every $a \in \mathcal{P}(y)$, $x_a(y)$ is positive odd and

$$C(x_a(y)) = y, \quad a(x_a(y)) = a.$$

Every positive odd predecessor of y occurs exactly once in this list.

Proof. Modulo three, $2^a y = 1$ is impossible when $3 \mid y$. Since 2^a is 1 for even a and 2 for odd a modulo three, the other two cases have the stated parities. The numerator is positive and odd; division by three preserves oddness. The identity $3x_a(y) + 1 = 2^a y$ with y odd proves both displayed first-return identities. Solving that same identity for x proves uniqueness. $\square$

Define

$$(Lf)(y) = u \sum_{a \in \mathcal{P}(y)} v^{a-1} f(x_a(y)). \quad (6)$$

This formula has several distinct exact domains.

Theorem 1.7 (algebraic, formal, and analytic transfer domains). *Let R be a commutative ring.*

- (i) *Formula (6) defines an $R[u, v]$ -linear endomorphism of the finite-support module $R[u, v]^{(\mathcal{O})}$.*
- (ii) *It defines a continuous endomorphism of*

$$\mathcal{M} = (R[[v]][[u]])^{\mathcal{O}}.$$

All v -series in this product are interpreted coefficientwise. The module $\mathcal{M}$ carries the complete linear, uniform u -adic topology with

$$F^k \mathcal{M} = (u^k R[[v]][[u]])^{\mathcal{O}} \quad (k \geq 0).$$

For a fixed coefficient of v^j , only $a \leq j+1$ can contribute, and L obeys $L(F^k \mathcal{M}) \subseteq F^{k+1} \mathcal{M}$. Consequently

$$(I - L)^{-1} = \sum_{n \geq 0} L^n$$

is well defined u -adically and

$$[(I - L)^{-1} \delta_x](y) = Z_{x \rightarrow y}(u, v). \quad (7)$$

- (iii) *Let $u, v \in \mathbb{C}$ and $u \neq 0$. On counting-measure sequence spaces, L is bounded on $\ell^1(\mathcal{O})$ exactly when $|v| \leq 1$, and then*

$$\|L\|_{\ell^1 \rightarrow \ell^1} = |u|.$$

For $1 < p \leq \infty$, let q be conjugate to p , with $q = 1$ when $p = \infty$. Then L is bounded on $\ell^p(\mathcal{O})$ exactly when $|v| < 1$, and

$$\|L\|_{\ell^p \rightarrow \ell^p} = |u| (1 - |v|^{2q})^{-1/q}. \quad (8)$$

The same strict condition and norm hold on $c_0(\mathcal{O})$.

If $u = 0$, the zero operator is defined for every finite v .

Proof. For (i), a finitely supported f has only finitely many states x with $f(x) \neq 0$, and each such state has one image $C(x)$. Thus the apparently infinite predecessor formula is a finite sum on this domain.

For (ii), the coefficient of v^j can receive a term only from $a - 1 \leq j$. This proves coefficientwise finiteness. The factor u proves the filtration statement and hence the Neumann identity. Starting from δ_x , each application of L follows the unique forward edge from the current odd state and multiplies by uv^{a-1} . Induction gives

$$(L^n \delta_x)(y) = \begin{cases} u^n v^{A_n(x)-n}, & C^n(x) = y, \\ 0, & C^n(x) \neq y, \end{cases}$$

which proves (7).

For (iii), the input coordinates split into disjoint fibres $C^{-1}(y)$. The absolute row weights over a fibre are either

$$|u|(1, |v|^2, |v|^4, \dots) \quad \text{or} \quad |u|(|v|, |v|^3, |v|^5, \dots),$$

or the empty sequence. For $1 < p \leq \infty$, Hölder's inequality gives the row-functional norm as the ℓ^q norm of this weight sequence. The first row is the larger one, and its norm is exactly the right side of (8). Finite truncations with phases aligned to the weights approach equality, so the upper bound is sharp. If $|v| \geq 1$, those same finite truncations have unbounded row-functional norm.

For $p = 1$, each input state occupies exactly one column and has column sum $|u||v|^{a(x)-1}$. All positive exponents occur by Lemma 1.6. The supremum is $|u|$ when $|v| \leq 1$ and is infinite when $|v| > 1$. Finally, for $|v| < 1$, the same row estimate maps c_0 to itself: split the series into finitely many exponents, whose predecessors tend to infinity with y , and a uniformly small geometric tail. Finite-support phase-aligned vectors prove sharpness. When $|v| \geq 1$, choose values tending slowly to zero along one predecessor fibre and align their phases; the defining row sum fails to be bounded. This proves the c_0 assertion. $\square$

At $u = v = 1$, the operator is therefore norm one on ℓ^1 , but it is unbounded on ℓ^∞ and is not defined on the constant function one by the raw infinite-sum formula. An arbitrary product of polynomial-valued functions is likewise not an algebraic domain for L : the missing v -adic completion is essential.

1.6 Prefix cocycles and forward affine intercepts

For a finite exponent word $w = a_1 \cdots a_n$, put

$$A_k(w) = a_1 + \cdots + a_k, \quad A(w) = A_n(w),$$

and define the finite prefix coordinate

$$\rho(w) = \sum_{k=1}^n 3^{k-1} 2^{-A_k(w)} \in \mathbb{Z}_3. \quad (9)$$

For the empty word, $A(\emptyset) = 0$ and $\rho(\emptyset) = 0$.

Proposition 1.8 (typed prefix cocycle and its exact fibres). *For exponent words w, z in chronological concatenation order,*

$$\rho(wz) = \rho(w) + 3^{|w|} 2^{-A(w)} \rho(z). \quad (10)$$

Let

$$P = \{(m, A) \in \mathbb{N}_0^2 : A \geq m\}, \quad \theta_{(m,A)}(r) = 3^m 2^{-A} r \quad (r \in \mathbb{Z}_3).$$

With addition on P and multiplication

$$(r, p) \star (s, q) = (r + \theta_p(s), p + q),$$

the semidirect product $\mathbb{Z}_3 \rtimes_\theta P$ is a monoid and

$$\chi_{\text{pre}}(w) = (\rho(w), (|w|, A(w)))$$

is an injective monoid homomorphism from positive-exponent words.

More explicitly, put $R(w) = 2^{A(w)} \rho(w) \in \mathbb{Z}$. If $n > 1$, then

$$a_n = \nu_2(R(w) - 3^{n-1}), \quad R(a_1 \cdots a_{n-1}) = \frac{R(w) - 3^{n-1}}{2^{a_n}}. \quad (11)$$

Together with $a_1 = A(w)$ when $n = 1$, this is a two-sided recursive inverse on the image. Every fibre of χ_{pre} is therefore a singleton.

Proof. Splitting the sum (9) after $|w|$ terms and factoring $3^{|w|}2^{-A(w)}$ from the remainder proves (10). The endomorphisms satisfy $\theta_p\theta_q = \theta_{p+q}$, so the displayed semidirect multiplication is associative with identity $(0, (0, 0))$; the cocycle identity proves multiplicativity of χ_{pre} .

Multiplying (9) by 2^A gives

$$R(w) = \sum_{k=1}^n 3^{k-1} 2^{A-A_k}.$$

The last term is 3^{n-1} . After subtracting it, the preceding term has exact 2-adic valuation a_n , while every earlier term is divisible by $2^{a_{n-1}+a_n}$. Hence no cancellation changes that valuation, proving the first formula in (11). Division gives the same integer for the prefix word. Recursion recovers all exponents, which proves injectivity and the inverse identities. $\square$

The acting object here is a monoid, not a group: $\theta_{(m,A)}$ is not surjective on $\mathbb{Z}_3$ when $m > 0$. In the coordinate pair (m, A) its multiplier is $3^m 2^{-A}$, not $3^m 2^{-(m+A)}$.

The preceding injectivity is a statement about exponent words. Erasing the source and target of a path is a different operation and has exact infinite fibres.

Proposition 1.9 (exact source fibres of an exponent word). *For a word $w = (a_1, \dots, a_n)$ with $n \geq 0$, put $A = A(w)$ and*

$$B_w = \sum_{i=1}^n 3^{n-i} 2^{a_1 + \cdots + a_{i-1}} \in \mathbb{Z}.$$

The word w is the exact successive valuation word of a positive odd source x if and only if

$$x \equiv 3^{-n}(2^A - B_w) \pmod{2^{A+1}}. \quad (12)$$

Thus every exponent word has one odd residue class of positive sources modulo 2^{A+1} , hence infinitely many path-category realizations. Retaining the source object selects one of them.

Proof. If $n = 0$, the claim is the tautology $x \equiv 1 \pmod{2}$ for a positive odd source, and the word is empty. Now let $n \geq 1$. We prove directly, by induction on n , that w is the exact valuation word of x if and only if

$$3^n x + B_w \equiv 2^A \pmod{2^{A+1}}. \quad (13)$$

For $n = 1$, this is $3x + 1 \equiv 2^{a_1} \pmod{2^{a_1+1}}$, exactly the assertion $\nu_2(3x + 1) = a_1$.

For $n > 1$, write $w' = (a_2, \dots, a_n)$, $A' = A - a_1$, and let $B_{w'}$ be its displayed integer. Then

$$B_w = 3^{n-1} + 2^{a_1} B_{w'}.$$

Reducing (13) modulo 2^{a_1} first shows that 2^{a_1} divides $3x + 1$. Put $x_1 = (3x + 1)/2^{a_1}$. Dividing the full congruence by 2^{a_1} gives

$$3^{n-1} x_1 + B_{w'} \equiv 2^{A'} \pmod{2^{A'+1}}.$$

Here $A' \geq 1$ and $B_{w'}$ is odd, so reduction modulo two shows that x_1 is odd; positivity is immediate from $x > 0$. By induction the displayed congruence is equivalent to x_1 being positive odd with exact valuation word w' . In particular x_1 is odd, so the first valuation is exactly a_1 . Reversing the argument proves the converse.

Solving (13) for x is legitimate because 3^n is a unit modulo 2^{A+1} , and gives the unique class (12). For $n \geq 1$, the integer B_w is odd: its first summand is odd and every later summand is even. Hence the class is odd; the $n = 0$ class was handled separately. Adding positive multiples of 2^{A+1} gives infinitely many positive sources. This direct induction also supplies, without using it as a proof dependency, the finite characterization appearing in Tao's source [7, source lines 455–470]. $\square$

Now put

$$g_a(X) = \frac{3X + 1}{2^a}$$

and apply a_1 first. The forward chronological composite is

$$G_w = g_{a_n} \circ \cdots \circ g_{a_1} = 3^n 2^{-A(w)} X + \beta(w), \quad (14)$$

where

$$\beta(w) = \sum_{i=1}^n 3^{n-i} 2^{-(a_i + \cdots + a_n)}. \quad (15)$$

Thus $B_w = 2^{A(w)} \beta(w)$, now as a consequence of the two displayed formulas rather than as a definition used before β is introduced.

Proposition 1.10 (exact reversal to Tao's offset). *For every finite positive-exponent word,*

$$\beta(w) = \rho(w^{\text{rev}}).$$

Moreover $\beta(w)$ is exactly Tao's finite offset $F_n(a_1, \dots, a_n)$ in the forward convention of [7, source lines 208–229 and 384–393]. Thus the prefix coordinate and forward intercept are connected by word reversal; they are not equal on the same chronological word in general.

Proof. Reverse the index in (9). The k th prefix sum of w^{rev} is the k -term suffix sum of w , and its coefficient 3^{k-1} becomes 3^{n-i} . This is exactly (15). Induction in (14) gives the same sum, and direct comparison with Tao's displayed definition gives the final identity. No external reversal assertion is used: the displayed reindexing is the complete comparison map. $\square$

For example,

$$\rho(1, 4) = \rho(3, 1, 1) = \frac{19}{32}, \quad \beta(1, 4) = \frac{5}{32}, \quad G_{(1,4)}(X) = \frac{9X + 5}{32}.$$

The distinction is not cosmetic: $G_{(1,4)}(3) = 1$, whereas the unreversed prefix constant would give the wrong value. The first equality also shows that ρ alone is not injective. The decoration in χ_{pre} is essential: the two words have respective $(|w|, A(w))$ coordinates $(2, 5)$ and $(3, 5)$.

Because target-left category multiplication writes later arrows on the left, χ_{pre} becomes covariant on the path algebra only after its coefficient monoid is replaced by the opposite monoid. This typing is made explicit in Proposition 1.23 below.

1.7 Gaussian prefix angles and the Collatz–Machin relation lattice

The individual summands of the prefix cocycle carry a second, multiplicative structure. It is independent of whether the exponent word is realized by a particular positive source; Proposition 1.9 then shows that every word considered below has infinitely many such sources.

Definition 1.11 (Collatz–Machin relation lattice). Let $w = (a_1, \dots, a_n)$ be a nonempty positive exponent word. For $1 \leq k \leq n$, define

$$r_k(w) = \frac{3^{k-1}}{2^{A_k(w)}}, \quad \vartheta_k(w) = \arctan r_k(w), \quad z_k(w) = 2^{A_k(w)} + i3^{k-1}, \quad (16)$$

where $\arctan$ is the principal value in $(0, \pi/2)$, and put

$$u_k(w) = \frac{z_k(w)}{\overline{z_k(w)}} = \frac{1 + ir_k(w)}{1 - ir_k(w)} = e^{2i\vartheta_k(w)}. \quad (17)$$

The integer relation lattice is

$$\text{Rel}_\pi(w) = \left\{ m \in \mathbb{Z}^n : \sum_{k=1}^n m_k \vartheta_k(w) \in \pi\mathbb{Q} \right\}. \quad (18)$$

Write

$$\mathcal{U}_1(\mathbb{Q}(i)) = \{x \in \mathbb{Q}(i)^\times : x\bar{x} = 1\}, \quad \mu_4 = \{1, -1, i, -i\}.$$

Exponentiating (18) gives the exact criterion

$$m \in \text{Rel}_\pi(w) \iff \prod_{k=1}^n u_k(w)^{m_k} \in \mu_4. \quad (19)$$

Indeed the product lies in $\mathbb{Q}(i)^\times$, whose roots of unity are exactly μ_4 ; conversely a value in μ_4 places the angle sum in $(\pi/4)\mathbb{Z}$. This is a multiplicative torsion test, not an additive unit equation.

The fixed-instance theory is classical. Conway, Radin, and Sadun reconstruct the $d = 1$ Størmer theory by orienting the Gaussian primes above $p \equiv 1 \pmod{4}$ and prove the resulting prime angles independent modulo π [15, Theorems 1–2 and Section 2]. Calcut’s Main Lemma and Corollaries 1–3 give the same μ_4 restriction and exclude a rational multiple of π for a single rational tangent other than $0, \pm 1$ [16]. We now type that factorization as an exact morphism for the present prefix family.

Theorem 1.12 (Gaussian coordinates for every fixed word). *For each rational prime $p \equiv 1 \pmod{4}$, choose one Gaussian prime ϖ_p above p , so $p = \varpi_p \overline{\varpi_p}$. There is an isomorphism*

$$\nu_G : \mathcal{U}_1(\mathbb{Q}(i))/\mu_4 \xrightarrow{\sim} \bigoplus_{p \equiv 1 \pmod{4}} \mathbb{Z}, \quad [x] \mapsto (v_{\varpi_p}(x))_p, \quad (20)$$

with inverse

$$(e_p)_p \mapsto \left[\prod_p \left(\frac{\varpi_p}{\overline{\varpi_p}} \right)^{e_p} \right].$$

For the word w , define

$$E_w : \mathbb{Z}^n \longrightarrow \bigoplus_{p \equiv 1 \pmod{4}} \mathbb{Z}, \quad (E_w)_{p,k} = v_{\varpi_p}(z_k(w)) - v_{\overline{\varpi_p}}(z_k(w)). \quad (21)$$

Then

$$\text{Rel}_\pi(w) = \ker E_w. \quad (22)$$

Every fibre over $y \in \text{im } E_w$ is the coset $m + \text{Rel}_\pi(w)$ for any m with $E_w(m) = y$, and the induced map

$$\mathbb{Z}^n / \text{Rel}_\pi(w) \xrightarrow{\sim} \text{im } E_w$$

has inverse $y \mapsto m + \text{Rel}_\pi(w)$. Changing the chosen orientation over one split prime changes only the sign of that coordinate. Every fibre over a point outside $\text{im } E_w$ is empty.

Proof. Unique factorization in $\mathbb{Z}[i]$ gives a finite prime factorization for every $x \in \mathbb{Q}(i)^\times$. If $x\bar{x} = 1$, the valuations at the two primes above a split p are opposite. A prime $p \equiv 3 \pmod{4}$ is fixed by conjugation, and $1+i$ is associated to its conjugate, so their valuations must be zero. The remaining factor is a Gaussian unit. Quotienting by μ_4 therefore leaves exactly one integer exponent for each oriented split prime. This proves that (20) is well defined, injective, and surjective, and the displayed product proves its inverse formula. All coordinate vectors have finite support.

Define

$$\bar{\Gamma}_w : \mathbb{Z}^n \longrightarrow \mathcal{U}_1(\mathbb{Q}(i))/\mu_4, \quad m \longmapsto \left[\prod_{k=1}^n u_k(w)^{m_k} \right].$$

It is a homomorphism, and (19) says that its kernel is $\text{Rel}_\pi(w)$. Since

$$v_{\varpi_p}(u_k(w)) = v_{\varpi_p}(z_k(w)) - v_{\varpi_p}(\overline{z_k(w)}) = v_{\varpi_p}(z_k(w)) - v_{\overline{\varpi_p}}(z_k(w)),$$

the composite $\nu_G \circ \bar{\Gamma}_w$ is exactly E_w . The kernel formula follows because ν_G is injective. The fibre and inverse-on-image statements are the first isomorphism theorem written in coordinates; they introduce no choice after passage to the quotient coset. $\square$

The theorem makes every fixed word decidable after exact factorization of the finitely many norms

$$N(z_k(w)) = 2^{2A_k(w)} + 3^{2(k-1)}.$$

Its content is not the conjecture below: the latter asks for a uniform rank classification as both the word and these factorizations vary.

Theorem 1.13 (unweighted Collatz–Machin non-torsion). *For every nonempty positive exponent word $w = (a_1, \dots, a_n)$,*

$$\sum_{k=1}^n \vartheta_k(w) \notin \pi\mathbb{Q}.$$

Equivalently, $(1, \dots, 1) \notin \text{Rel}_\pi(w)$.

Proof. Put

$$Z(w) = \prod_{k=1}^n z_k(w) = X + iY \in \mathbb{Z}[i].$$

Modulo three, every factor with $k \geq 2$ is the nonzero real number $2^{A_k(w)}$, while the first factor is $2^{A_1(w)} + i$. Hence both X and Y are nonzero modulo three. It follows that $Z/\bar{Z}$ is neither 1 nor -1 , since those two equalities would force respectively $Y = 0$ or $X = 0$.

Modulo two, every factor is i , because $A_k(w) \geq 1$ and 3^{k-1} is odd. Thus $Z \equiv i^n \pmod{2}$: X and Y have opposite parity. The equalities $Z/\overline{Z} = i$ and $Z/\overline{Z} = -i$ would force $X = Y$ and $X = -Y$, respectively, both impossible for integers of opposite parity. Therefore $Z/\overline{Z} \notin \mu_4$. But

$$\frac{Z}{\overline{Z}} = \prod_{k=1}^n u_k(w),$$

so (19) completes the proof. $\square$

The coefficients cannot be removed from the statement: the legal word $(1, 1)$ has a genuine weighted relation.

Theorem 1.14 (the exact exceptional two-letter lattice). *For $w = (1, 1)$,*

$$\text{Rel}_\pi(1, 1) = \mathbb{Z}(2, 1), \quad 2 \arctan(1/2) + \arctan(3/4) = \frac{\pi}{2}. \quad (23)$$

Proof. Here

$$z_1 = 2 + i, \quad z_2 = 4 + 3i = i(2 - i)^2 = i\overline{z_1}^2,$$

and therefore

$$(2 + i)^2(4 + 3i) = 25i, \quad u_2 = -u_1^{-2}.$$

It follows that $u_1^2 u_2 = -1$ and hence $(2, 1)$ belongs to the relation lattice. Since both angles lie in $(0, \pi/4)$, the resulting congruence is the displayed equality with $\pi/2$, not merely equality modulo π .

The Gaussian prime $2 + i$ divides the numerator but not the denominator of $u_1 = (2 + i)/(2 - i)$, so $v_{2+i}(u_1^d) = d$. Every element of μ_4 has $v_{2+i} = 0$; hence $u_1^d \in \mu_4$ forces $d = 0$. For arbitrary (m_1, m_2) ,

$$u_1^{m_1} u_2^{m_2} = (-1)^{m_2} u_1^{m_1 - 2m_2}$$

lies in μ_4 exactly when $m_1 - 2m_2 = 0$. Hence the full lattice, including all zero and nonzero right-hand sides, is $\mathbb{Z}(2, 1)$. $\square$

The same identity occurs in Gasull, Luca, and Varona as

$$\arctan(1/2) + \frac{1}{2} \arctan(3/4) = \pi/4$$

inside their classification of two-term, nonzero-right-hand-side formulas with 2-integer arguments [17, Theorem 1]. Their theorem does not state the full higher-length prefix classification.

Theorem 1.15 (exact full-grid relation lattice). *Let*

$$\mathcal{G}_{15,60} = \{(k, A) : 1 \leq k \leq 15, k \leq A \leq 60\}, \quad |\mathcal{G}_{15,60}| = 795,$$

and attach $z_{k,A} = 2^A + i3^{k-1}$ and $\vartheta_{k,A} = \arctan(3^{k-1}/2^A)$ to each column. Define

$$L_{15,60} := \text{Rel}_\pi(\mathcal{G}_{15,60}) = \left\{ m \in \mathbb{Z}^{\mathcal{G}_{15,60}} : \sum_{(k,A) \in \mathcal{G}_{15,60}} m_{k,A} \vartheta_{k,A} \in \pi\mathbb{Q} \right\}.$$

The complete integer relation lattice on these 795 angles is

$$L_{15,60} = \mathbb{Z}\mathbf{r}_1 \oplus \mathbb{Z}\mathbf{r}_2, \quad (24)$$

where

$$\begin{aligned}\mathbf{r}_1 &= 2e_{(1,1)} + e_{(2,2)}, \\ \mathbf{r}_2 &= 2e_{(1,1)} + e_{(1,2)} + e_{(1,3)} + e_{(3,5)}.\end{aligned}$$

The two exact Gaussian certificates are

$$(2+i)^2(4+3i) = 25i$$

and

$$(2+i)^2(4+i)(8+i)(32+9i) = 5525i.$$

Consequently every positive word with $n \leq 15$ and $A_n(w) \leq 60$ has exactly the lattice predicted in Conjecture 1.16; there is no bound on the coefficients of a tested relation.

Proof. The portable certificate `certificates/arctangent_relation_lattice_checks.py` uses only exact integer and finite-field arithmetic. It takes all 1,008 primes $q \leq 38,821$ with $q \equiv 1 \pmod{12}$. A prime is retained when every $z_{k,A}/\overline{z_{k,A}}$ has good reduction in $\mathbb{F}_q$. For each of the 794 retained primes, choose a square root of -1 and apply the cubic character

$$\chi_q(x) = x^{(q-1)/3}.$$

Because 4 divides $(q-1)/3$, this character kills μ_4 . After identifying its values with $\mathbb{F}_3$, every genuine relation therefore reduces to a vector in the kernel of the resulting 794×795 matrix over $\mathbb{F}_3$. Exact row reduction gives rank 793 and nullity two. The rank before the last prime is 792; 38,821 is the first prime giving full rank. The reductions of $\mathbf{r}_1$ and $\mathbf{r}_2$ are independent kernel vectors, so they are the complete mod-three kernel.

Let

$$\overline{\Gamma}_{\mathcal{G}} : \mathbb{Z}^{\mathcal{G}^{15,60}} \longrightarrow \mathcal{U}_1(\mathbb{Q}(i))/\mu_4$$

be the product map defined as in Theorem 1.12. Let L be its kernel and $\mathcal{L}_0 = \mathbb{Z}\mathbf{r}_1 + \mathbb{Z}\mathbf{r}_2$. The displayed Gaussian products prove $\mathcal{L}_0 \subseteq L$. The character calculation gives, for every $\ell \in L$, an expression $\ell = r + 3z$ with $r \in \mathcal{L}_0$ and $z \in \mathbb{Z}^{\mathcal{G}^{15,60}}$. In the target $\mathcal{U}_1(\mathbb{Q}(i))/\mu_4$ this implies $\overline{\Gamma}_{\mathcal{G}}(z)^3 = 1$. That target is torsion-free: if $[x]^d = 1$, then $x^d \in \mu_4$, hence $x^{4d} = 1$ and $x \in \mu_4$. Therefore $z \in L$ and $L = \mathcal{L}_0 + 3L$.

On the two coordinates $(2,2)$ and $(1,2)$, the columns of $(\mathbf{r}_1, \mathbf{r}_2)$ form the identity matrix. Thus $\mathcal{L}_0$ is a direct summand of the ambient integer lattice. The subgroup $L/\mathcal{L}_0$ is consequently finitely generated and free, while $L/\mathcal{L}_0 = 3(L/\mathcal{L}_0)$; reducing a free basis modulo three forces $L/\mathcal{L}_0 = 0$. This proves (24) over the integers.

A prefix chain contains at most one column for each k . In $a\mathbf{r}_1 + b\mathbf{r}_2$, the exact coefficients at both $(1,2)$ and $(1,3)$ equal b . Thus $b \neq 0$ is incompatible with support on a prefix chain. A nonzero multiple of $\mathbf{r}_1$ is supported on a chain exactly when that chain contains $(1,1)$ and $(2,2)$, equivalently when $(a_1, a_2) = (1,1)$. The coordinate embedding $e_k \mapsto e_{(k, A_k(w))}$ identifies the word relation lattice with the intersection of $L_{15,60}$ and its chain coordinate subspace. This proves the final assertion. $\square$

The certificate pins the grid, prime lists, full trit matrix, independent-row list, and pivot columns by SHA-256. It also checks all 631,230 character entries, both Gaussian products, the unit saturation minor, and all nine mod-three combinations of the script vectors named **R1**, **R2**, which are $\mathbf{r}_1, \mathbf{r}_2$ above. It certifies no point outside the displayed grid.

Conjecture 1.16 (Collatz–Machin rigidity). *For every nonempty positive exponent word $w = (a_1, \dots, a_n)$,*

$$\text{Rel}_\pi(w) = \begin{cases} \mathbb{Z}(2, 1, 0, \dots, 0), & n \geq 2 \text{ and } (a_1, a_2) = (1, 1), \\ \{0\}, & \text{otherwise.} \end{cases}$$

The fixed-word theorem proves that this is exactly a uniform signed-valuation rank statement for the Gaussian integers $2^{A_k(w)} + i3^{k-1}$. Theorems 1.13 and 1.14 prove two all-length boundaries, and Theorem 1.15 proves the complete result for $n \leq 15$, $A_n \leq 60$. Conway–Radin–Sadun, Section 2, supplies the classical Gaussian-UFD coordinates for fixed rational angles; Gasull–Luca–Varona, Theorem 1, supplies its stated two-term classification and the $\mathbf{r}_1$ identity. Neither cited statement establishes the 795-column lattice or the uniform prefix-word classification. This is a source-scope comparison, not a novelty or priority claim. The conjecture concerns relations among prefix angles; it asserts neither Collatz termination nor any endpoint property of an orbit.

1.8 Finite residue transformation semigroups

Fix $N \geq 1$ and write

$$R_N = \mathbb{Z}/3^N\mathbb{Z}, \quad U_N = R_N^\times, \quad \varphi_N = |U_N| = 2 \cdot 3^{N-1}.$$

Since

$$\text{ord}_{3^N}(2) = 2 \cdot 3^{N-1},$$

the elements 2^{-a} , $a \geq 1$, run through all of U_N . For $\eta \in U_N$ put

$$G_\eta(r) = \eta(3r + 1) = 3\eta r + \eta.$$

Let S_N be the image of all nonempty words in the transformations G_η , and let

$$T_N = S_N \sqcup \{\text{id}_{R_N}\}.$$

Every nonempty word sends 0 to a unit, so the identity is not in S_N . Thus S_N is a transformation semigroup and T_N is its identity-adjoined transformation monoid.

For chronological $\boldsymbol{\eta} = (\eta_1, \dots, \eta_k)$, with η_1 applied first, direct expansion gives

$$G_{\boldsymbol{\eta}}(r) = 3^k \left(\prod_{i=1}^k \eta_i \right) r + \sum_{j=1}^k 3^{k-j} \prod_{i=j}^k \eta_i. \quad (25)$$

The constant term is congruent to η_k modulo three and is therefore a unit.

Lemma 1.17 (order of two modulo powers of three). *For every $N \geq 1$, $\text{ord}_{3^N}(2) = 2 \cdot 3^{N-1}$.*

Proof. By Euler’s theorem the order divides $\varphi(3^N) = 2 \cdot 3^{N-1}$, and reduction modulo three shows that it is even. Hence it has the form $2 \cdot 3^j$ with $0 \leq j \leq N-1$. The lifting-the-exponent identity gives

$$\nu_3(2^{2 \cdot 3^j} - 1) = \nu_3(4^{3^j} - 1) = \nu_3(4 - 1) + \nu_3(3^j) = j + 1.$$

The congruence $2^{2 \cdot 3^j} \equiv 1 \pmod{3^N}$ therefore holds exactly when $j \geq N-1$. The only possible order is $2 \cdot 3^{N-1}$; the case $N = 1$ is included with $j = 0$. $\square$

For $N \geq 2$ define

$$L_{N,1} = \{r \mapsto 3br + b : b \in U_N\},$$

and for $2 \leq k < N$ define

$$L_{N,k} = \{r \mapsto 3^k sr + b : s \in (\mathbb{Z}/3^{N-k}\mathbb{Z})^\times, b \in U_N\}.$$

Finally let

$$\mathcal{C}_N = \{c_b : b \in U_N\}, \quad c_b(r) = b.$$

Theorem 1.18 (exact finite residue normal form). *For $N = 1$,*

$$S_1 = \mathcal{C}_1 = \{c_1, c_2\}.$$

For every $N \geq 2$, there is a disjoint union

$$S_N = L_{N,1} \sqcup \bigsqcup_{k=2}^{N-1} L_{N,k} \sqcup \mathcal{C}_N. \quad (26)$$

More precisely, $L_{N,k}$ is exactly the set represented by words of length $k < N$, every word of length at least N is constant, and every unit-valued constant already has a representative of length N .

Proof. Formula (25) shows that a word of length $k < N$ has slope of exact 3-adic valuation k and unit intercept; at length at least N it is constant. At length one, slope and intercept satisfy the exceptional relation $\alpha = 3b$, giving $L_{N,1}$.

We prove surjectivity onto every later stratum constructively. First suppose $N \geq 3$. At length two, prescribe $s \in (\mathbb{Z}/3^{N-2}\mathbb{Z})^\times$ and $b \in U_N$. Put

$$q = sb^{-1} \pmod{3^{N-2}}, \quad \eta_1 = \frac{q}{1-3q} \pmod{3^{N-2}},$$

choose any unit lift of η_1 to U_N , and set

$$\eta_2 = \frac{b}{1+3\eta_1} \pmod{3^N}.$$

Then

$$\eta_2(3\eta_1 + 1) = b, \quad \eta_2\eta_1 \equiv s \pmod{3^{N-2}},$$

so $G_{\eta_2} \circ G_{\eta_1}(r) = 9sr + b$. When $N = 2$, choose an arbitrary $\eta_1 \in U_2$ and put $\eta_2 = b/(1+3\eta_1)$ in U_2 ; the resulting length-two word is the constant c_b .

Suppose all length- $(k-1)$ forms occur, where $3 \leq k < N$. Given a desired $3^k s'r + b'$, choose any prior unit intercept b_0 , put

$$\eta_k = \frac{b'}{3b_0 + 1},$$

and choose the prior slope parameter s_0 so that $s_0 \equiv s'\eta_k^{-1} \pmod{3^{N-k}}$. Applying G_{η_k} on the left gives the prescribed slope and intercept. To obtain a prescribed constant $c_{b'}$ at length $N \geq 3$, choose any already constructed length- $(N-1)$ form with intercept b_0 and apply G_{η_N} with $\eta_N = b'/(3b_0 + 1)$; its slope vanishes modulo 3^N . Equality of two affine maps is equivalent to equality of their values at 0 and 1, hence of intercept and slope; distinct strata have different slope valuations. This proves both completeness and disjointness. $\square$

Corollary 1.19 (cardinalities and ranks). *For $N \geq 2$,*

$$|S_N| = \varphi_N(3^{N-2} + 1), \quad |T_N| = 1 + \varphi_N(3^{N-2} + 1). \quad (27)$$

The first six semigroup cardinalities are

$$2, 12, 72, 540, 4536, 39852,$$

and the corresponding monoid cardinalities are

$$3, 13, 73, 541, 4537, 39853.$$

An element of $L_{N,k}$ has image $b + 3^k R_N$ and rank 3^{N-k} ; the constants have rank one, while the adjoined identity has rank 3^N .

Proof. The stratum sizes are

$$|L_{N,1}| = \varphi_N, \quad |L_{N,k}| = \varphi_N \varphi_{N-k} \quad (2 \leq k < N), \quad |\mathcal{C}_N| = \varphi_N.$$

Since $\sum_{j=1}^{N-2} \varphi_j = 3^{N-2} - 1$, their sum is (27). The image and rank statements follow by varying $3^k sr$ over the ideal $3^k R_N$. $\square$

Proposition 1.20 (the minimum ideal). *The rank-one stratum $\mathcal{C}_N$ is the minimum two-sided ideal of both S_N and T_N . Its multiplication is left-zero:*

$$c_b c_d = c_b.$$

Proof. For every $f \in T_N$ and unit b ,

$$c_b f = c_b, \quad f c_b = c_{f(b)} \in \mathcal{C}_N,$$

because every transformation in T_N sends units to units. Hence $\mathcal{C}_N$ is a two-sided ideal. If a nonempty ideal contains f , then it contains $c_b f = c_b$ for every $b \in U_N$, so it contains $\mathcal{C}_N$. This direct proof agrees with the general minimum-rank description for faithful finite transformation monoids [10, Proposition 13.1 and Lemma 13.2]. $\square$

1.9 Reduction maps and the exact inverse limit

For $M > N$, coefficient reduction gives

$$\pi_{M,N} : S_M \longrightarrow S_N, \quad (\alpha, b) \longmapsto (\alpha \bmod 3^N, b \bmod 3^N).$$

Theorem 1.21 (surjectivity and exact reduction fibres). *The maps $\pi_{M,N}$ are surjective semigroup homomorphisms. Put $d = M - N$. If $N \geq 2$, then*

$$|\pi_{M,N}^{-1}(s)| = \begin{cases} 3^d, & s \in L_{N,1}, \\ 3^{2d}, & s \in L_{N,k}, \quad 2 \leq k < N, \\ 3^{2d}, & s \in \mathcal{C}_N. \end{cases} \quad (28)$$

For the bottom reduction $S_M \rightarrow S_1$, each of the two fibres has size

$$3^{M-1}(3^{M-2} + 1).$$

The maps extend to $T_M \rightarrow T_N$, and the identity has the singleton fibre $\{\text{id}\}$.

Proof. Reduction intertwines every generator and therefore every word. The normal form theorem supplies a lift, proving surjectivity. A target in $L_{N,1}$ has exactly the 3^d unit lifts of its intercept; its slope remains forced to be three times that intercept. In an intermediate stratum, the intercept and unit slope parameter each have 3^d lifts, giving 3^{2d} .

A constant target receives 3^d constant lifts and lifts from every source stratum of slope valuation $k \geq N$. Their number is

$$3^d \left(1 + \sum_{k=N}^{M-1} \varphi_{M-k} \right) = 3^d \left(1 + \sum_{j=1}^d \varphi_j \right) = 3^{2d}.$$

At level one both target elements are constants. Symmetry of the unit intercepts modulo three splits $|S_M|$ equally, giving the stated bottom fibre. A nonempty word has unit intercept, so it cannot reduce to the identity, which proves the last assertion. $\square$

Theorem 1.22 (exact 3-adic inverse limit). *Under coefficientwise reduction,*

$$\varprojlim_N S_N \cong \{x \mapsto 3bx + b : b \in \mathbb{Z}_3^\times\} \sqcup \{x \mapsto \alpha x + b : \alpha \in 9\mathbb{Z}_3, b \in \mathbb{Z}_3^\times\}. \quad (29)$$

The second family includes $\alpha = 0$. Moreover

$$\varprojlim_N T_N = \{\text{id}_{\mathbb{Z}_3}\} \sqcup \varprojlim_N S_N.$$

For $\eta \in \mathbb{Z}_3^\times$ define $G_\eta(x) = \eta(3x + 1)$, and distinguish the two semigroups

$$\mathcal{S}_{\text{unit}} = \langle G_\eta : \eta \in \mathbb{Z}_3^\times \rangle, \quad \mathcal{S}_{\text{exp}} = \langle G_{2^{-a}} : a \geq 1 \rangle.$$

In the coefficientwise 3-adic topology,

$$\overline{\mathcal{S}_{\text{exp}}} = \overline{\mathcal{S}_{\text{unit}}} = \varprojlim_N S_N.$$

Proof. A compatible transformation family determines unique compatible coefficients

$$b_N = f_N(0), \quad \alpha_N = f_N(1) - f_N(0),$$

hence unique $b \in \mathbb{Z}_3^\times$ and $\alpha \in \mathbb{Z}_3$. At every level $N \geq 2$, the normal form says either $\alpha_N = 3b_N$ or $\alpha_N \in 9\mathbb{Z}_3$. A higher-level element of the second kind cannot reduce to a level- N element of the first kind, whose slope has valuation exactly one. Thus a compatible family lies entirely in the first branch from level two onward, giving $\alpha = 3b$, or entirely in the second, giving $\alpha \in 9\mathbb{Z}_3$. Conversely every displayed coefficient pair reduces to the finite normal form at every level, so it defines a compatible family. Composition of affine coefficient pairs is

$$(\alpha, b) \circ (\gamma, d) = (\alpha\gamma, \alpha d + b).$$

In the displayed family b, d are units and $3 \mid \alpha$, so $\alpha d + b$ is a unit, while $\alpha\gamma \in 9\mathbb{Z}_3$. Hence the family is closed under composition, and coefficient extraction respects composition. This proves the asserted semigroup isomorphism. The identity-fibre statement in the preceding theorem prevents a mixed identity sequence.

For arbitrary unit generators, the constructive normal-form proof works unchanged over $\mathbb{Z}_3$: a word of length one gives $\alpha = 3b$, and words of each fixed length $k \geq 2$ give every pair with

$\nu_3(\alpha) = k$ and unit intercept. Closing these strata adds exactly the zero slopes. Thus $\overline{\mathcal{S}_{\text{unit}}}$ is the displayed inverse limit.

For the actual exponent generators, Lemma 1.17 shows that their reductions 2^{-a} fill U_N at every level N . Given any cylinder determined by an element of S_N , choose an exponent separately for every unit letter in one representing word. The resulting element of $\mathcal{S}_{\text{exp}}$ has the prescribed reduction. Therefore $\mathcal{S}_{\text{exp}}$ meets every finite-level cylinder and is dense in the same inverse limit. This proves both closure equalities without identifying the countable exponent semigroup with the all-unit semigroup. $\square$

The slope condition cannot be weakened to $\alpha \in 3\mathbb{Z}_3$ with arbitrary unit intercept. At valuation one the exact relation is $\alpha = 3b$; for example $x \mapsto 3x + 2$ is already excluded modulo nine.

1.10 Typed finite actions and the two source obstructions

The specific object-erasing assignment that sends each path to its residue transformation cannot extend linearly to an algebra representation in one transformation algebra. For distinct objects $x \neq y$,

$$e_x e_y = 0,$$

whereas erasing the object coordinate sends both empty paths to the same identity transformation. The product of their images is then nonzero.

There is a separate variance defect. For a map $h : R_N \rightarrow R_N$, let $P_h f = f \circ h$. Then

$$P_h P_k = P_{k \circ h},$$

so pullback reverses composition. This is visible on the actual composable path

$$3 \xrightarrow{1} 5 \xrightarrow{4} 1.$$

Modulo nine,

$$G_1(r) = 6r + 5, \quad G_4(r) = 3r + 4,$$

but

$$G_4 \circ G_1 = c_1, \quad G_1 \circ G_4 = c_2.$$

Thus the two orderings are genuinely different transformations, not merely different notation.

Let $R_0 = R[u, v]$ and use the monoid algebra $R_0[T_N]$ with multiplication induced by ordinary target-left transformation composition. Put

$$\mathcal{B}_N = \text{Mat}_{\mathcal{O}}^{\text{fin}}(R_0[T_N]),$$

the algebra of matrices having finite support in the infinite object index. For the decorated prefix coordinate, put

$$\mathcal{B}_{\text{pre}} = \text{Mat}_{\mathcal{O}}^{\text{fin}}(R_0[(\mathbb{Z}_3 \rtimes_{\theta} P)^{\text{op}}]),$$

where the bracket denotes the finite-support monoid algebra.

Proposition 1.23 (typed forward and pullback representations). *For $p : x \rightarrow y$, let G_p be its forward residue transformation. The map*

$$\Pi_N([p]) = w(p)E_{y,x} \otimes [G_p] \tag{30}$$

is an R_0 -algebra homomorphism $\mathfrak{A}_C \rightarrow \mathcal{B}_N$.

On the explicitly based free module

$$\mathcal{V}_N = \bigoplus_{x \in \mathcal{O}} \bigoplus_{r \in R_N} R_0 e_{x,r}$$

it has the concrete forward action

$$\Pi_N([p])e_{x,r} = w(p)e_{y,G_p(r)},$$

and is zero on every other object fibre.

For pullback, put

$$\mathcal{F}_N = \text{Fun}(R_N, R_0), \quad \mathcal{K}_N = \text{Mat}_{\mathcal{O}}^{\text{fin}}(\text{End}_{R_0}(\mathcal{F}_N)).$$

The typed pullback map

$$K_N([p]) = w(p)E_{x,y} \otimes P_{G_p}$$

from $\mathfrak{A}_C$ to $\mathcal{K}_N$ is an anti-homomorphism:

$$K_N(ab) = K_N(b)K_N(a).$$

Finally, the decorated prefix map has the covariant typed realization

$$\Psi([p]) = w(p)E_{t(p),s(p)} \otimes [\chi_{\text{pre}}(\text{word}(p))]_{\text{op}}$$

as an R_0 -algebra homomorphism $\Psi : \mathfrak{A}_C \rightarrow \mathcal{B}_{\text{pre}}$.

Proof. If $p : x \rightarrow y$ and $q : z \rightarrow x$, then matrix-unit multiplication and forward composition give

$$\Pi_N([p])\Pi_N([q]) = w(p)w(q)E_{y,z} \otimes [G_p \circ G_q] = \Pi_N([p \circ q]).$$

For noncomposable paths the matrix units multiply to zero. Also $\Pi_N(e_x) = E_{x,x} \otimes [\text{id}]$, so the object idempotents remain orthogonal. This proves the algebra and concrete-module statements.

The identity $P_h P_k = P_{k \circ h}$ proves the pullback anti-homomorphism, with the transposed matrix unit sending target functions back to source functions. For the last statement, the chronological word of $p \circ q$ is the word of q followed by the word of p . Proposition 1.8 therefore gives

$$\chi_{\text{pre}}(\text{word}(p \circ q)) = \chi_{\text{pre}}(\text{word}(q)) \star \chi_{\text{pre}}(\text{word}(p)),$$

which is exactly the product in the opposite semigroup algebra after the matrix units have enforced composability. $\square$

Theorem 1.24 (exact kernels of the typed representations). *Let R be a nonzero commutative ring. The decorated prefix representation Ψ is injective. For $F = \sum_p f_p[p] \in \mathfrak{A}_C$, the finite-residue representation has the exact kernel criterion*

$$F \in \ker \Pi_N \iff \sum_{\substack{p: x \rightarrow y \\ G_p = h}} f_p w(p) = 0 \quad \text{for every } x, y \in \mathcal{O} \text{ and } h \in T_N. \quad (31)$$

In particular, every Π_N is noninjective. If $\epsilon_1 : 1 \rightarrow 1$ is the exponent-two loop, then

$$uv[\epsilon_1^N] - [\epsilon_1^{N+1}] \in \ker \Pi_N \cap \ker K_N \quad (32)$$

is nonzero.

Proof. Distinct paths have either different source-target matrix units or different exponent words. In the latter case the decorated map χ_{pre} separates them by Proposition 1.8. Hence distinct path basis elements have distinct matrix-monoid basis coordinates under Ψ . Multiplication by the monomial $w(p)$ only shifts polynomial exponents and is injective in $R[u, v]$. Finite linear independence proves that Ψ is injective.

For Π_N , collect the image of a finite sum by its matrix coordinate (y, x) and transformation basis element $[h]$. Its coefficient is exactly the sum on the right of (31); independence of those basis coordinates proves both directions of the criterion.

Finally $a(1) = 2$, so $w(\epsilon_1) = uv$. In R_N , where 4 is invertible, the residue transformation $H = G_{\epsilon_1}$ satisfies

$$H(r) - 1 = \frac{3}{4}(r - 1).$$

Therefore $H^N(r) - 1 = (3/4)^N(r - 1) = 0$ in R_N , so $H^N = H^{N+1} = c_1$. The two terms in (32) consequently have the same image $(uv)^{N+1}E_{1,1} \otimes [c_1]$. Their pullback images likewise agree, because both use P_{c_1} and the same matrix unit. The original paths are distinct basis elements and R is nonzero, so their displayed difference is nonzero. No equality between the full kernels of K_N and Π_N is asserted. $\square$

The target $\mathcal{B}_N$ is finite in each residue fibre but not finite-dimensional: it retains the infinite object set $\mathcal{O}$. A genuinely finite quotient would require a separately proved categorical congruence. Merely reducing object labels modulo 3^N identifies orthogonal idempotents and does not provide one. Theorem 1.18 classifies the image transformations and their strata, not the word fibres. The exact linear collision criterion is instead (31).

1.11 Exact boundary of the construction

Several proposed extensions fail before any Collatz-specific issue arises. The exact failures determine which algebraic structures have actually been built.

Source repair 1.25 (four algebraic type failures). First, the natural reduction maps

$$\mathbb{Z}/3^{N+1}\mathbb{Z} \longrightarrow \mathbb{Z}/3^N\mathbb{Z}$$

form an inverse system, not the direct system printed in an early draft. Its limit is $\mathbb{Z}_3 = \varprojlim_N \mathbb{Z}/3^N\mathbb{Z}$; a residue class modulo 3^m has no canonical lift modulo 3^{m+n} .

Second, $\mathbb{R}_{>0}$ under its usual addition and multiplication is not a semiring because it has no additive zero. The path weights may instead take values in the multiplicative monoid $\mathbb{R}_{>0}$ or in the semiring $\mathbb{R}_{\geq 0}$.

Third, ordinary deconcatenation is a coalgebra coproduct on words but is not multiplicative for ordinary concatenation. For two letters a, b ,

$$\Delta(a)\Delta(b) = 1 \otimes ab + a \otimes b + b \otimes a + ab \otimes 1,$$

whereas

$$\Delta(ab) = 1 \otimes ab + a \otimes b + ab \otimes 1.$$

The uncanceled term $b \otimes a$ proves that the printed data do not define a bialgebra. An unshuffle coproduct, a braided tensor product, or another explicit compatibility law would be a different construction.

Fourth, the two-element set of local affine formulas $\{X/2, (3X + 1)/2\}$ is not closed under affine composition: already $(X/2) \circ (X/2) = X/4$. It is therefore not an algebra acting on the path algebra, and no crossed product with it is defined by the passage.

At the proved scope, the programme supplies:

- (1) an exact equivalence between exponent paths and legal local odd-to-odd blocks;
- (2) a genuine two-variable path weight and two explicitly compared $\mathbb{F}_2^2$ degrees;
- (3) endpoint formal series, a coefficientwise path completion, and exact resolvent identities;
- (4) a predecessor transfer operator with precise algebraic, formal, and analytic domains;
- (5) an injective decorated prefix map χ_{pre} and its exact reversal relation to the forward affine/Tao offset;
- (6) an exact Gaussian signed-valuation map for every prefix-angle relation lattice, an all-word unweighted non-torsion theorem, the exceptional two-letter lattice, and a coefficient-unbounded full-grid calculation;
- (7) finite residue transformation semigroups with complete normal forms, cardinalities, ranks, minimum ideals, reduction fibres, and inverse limit;
- (8) typed forward representations and pullback anti-representations that retain object idempotents and composition order.

Exact boundary 1.26. The displayed data do not construct a path groupoid of local homeomorphisms, a groupoid C^* -algebra, a Leavitt or Cuntz–Krieger algebra, a quotient category QGr , a Hopf or bialgebra structure, a crossed coefficient product, or an operator-algebra completion. Names and analogies do not supply the absent objects or maps. None of the theorems in this paper proves termination of every positive Collatz orbit.

The portable certificate `certificates/finite_actions_checks.py` verifies bounded instances of the block inverse, grading fibres, cocycle and reversal formulas, the noninjective undecorated ρ example, source congruences, predecessor fibres, endpoint coefficients, finite path-cut counts, the exact generated residue semigroups through $N = 6$, their ranks and adjacent reduction fibres, finite projections of the inverse-limit classification, typed path products, and the explicit fixed-point-loop kernel relation. It has no private-corpus dependency. Source-identity pins belong to the nonportable critical workspace and are deliberately not copied into this package. The finite certificate does not independently prove the infinite, analytic, faithfulness, or exact-kernel theorems printed above.

The separate portable certificate `certificates/arctangent_relation_lattice_checks.py` proves the complete integer relation lattice on the 795-column grid $1 \leq k \leq 15$, $k \leq A \leq 60$ by exact cubic-character rows, finite-field rank, a saturated-lattice lift, and two Gaussian product identities. It has no coefficient search bound. It does not certify any column outside that grid or the all-length conjecture.

2 Two-adic survivor coding and exact interface boundaries

This section continues the separately authored theorem package of Section 1. Private chronological notes supplied questions and candidate interfaces, but they are not theorem authority. Every

Collatz-specific assertion retained below is proved from the displayed definitions. Public sources enter only for the exact objects and functoriality that they themselves construct.

The finite source classes of Proposition 1.9 assemble into an exact inverse-limit survivor coding. This is not the metric completion of the positive odd integers: those integers meet every odd residue class modulo every power of two, so their 2-adic completion is all of $\mathcal{O}_2 = 1 + 2\mathbb{Z}_2$. The dynamically coded domain is instead the survivor subspace obtained by removing $-1/3$ and every point whose forward orbit reaches it. We identify the positive-integer itineraries inside that survivor coding, compute an elementary full-shift pressure, and prove two typing obstructions for proposed tropical and prismatic interfaces.

2.1 The partial odd map and its exceptional preimage tree

Put

$$\mathcal{O}_2 = 1 + 2\mathbb{Z}_2, \quad x_* = -\frac{1}{3} \in \mathcal{O}_2.$$

The inclusion $x_* \in \mathcal{O}_2$ is literal: 3 is a unit of $\mathbb{Z}_2$ and $-1/3 \equiv 1 \pmod{2}$.

Definition 2.1 (survivor space). For $x \in \mathcal{O}_2 \setminus \{x_*\}$ define

$$a_2(x) = \nu_2(3x + 1) \in \mathbb{N}_{>0}, \quad C_2(x) = \frac{3x + 1}{2^{a_2(x)}} \in \mathcal{O}_2.$$

Set $E_0 = \{x_*\}$. For $n \geq 1$, let E_n be the set of $x \in \mathcal{O}_2$ for which $C_2^j(x)$ is defined for $0 \leq j < n$ and $C_2^n(x) = x_*$. Finally put

$$E = \bigcup_{n \geq 0} E_n, \quad X_\infty = \mathcal{O}_2 \setminus E. \quad (33)$$

The set X_∞ is exactly the domain of infinite accelerated itineraries. The exceptional set is not exhausted by the single point x_* .

Proposition 2.2 (typed inverse branches and the exceptional tree). For $a \in \mathbb{N}_{>0}$ and $y \in \mathcal{O}_2$, define

$$b_a(y) = \frac{2^a y - 1}{3}.$$

Then $b_a(y) \in \mathcal{O}_2$,

$$a_2(b_a(y)) = a, \quad C_2(b_a(y)) = y,$$

and $b_a(y)$ is the unique point with these two properties. Consequently E_n is countable for every n , E is countable, and $C_2(X_\infty) \subseteq X_\infty$.

The first exceptional preimages are

$$x_a = b_a(x_*) = -\frac{2^a + 3}{9}, \quad \nu_2(3x_a + 1) = a, \quad C_2(x_a) = -\frac{1}{3}. \quad (34)$$

Proof. Since $2^a y$ is even, $2^a y - 1$ is odd; division by the odd unit 3 keeps the result in $\mathcal{O}_2$. Direct substitution gives

$$3b_a(y) + 1 = 2^a y.$$

The factor y is a 2-adic unit, so the valuation is exactly a and the accelerated quotient is y . Conversely, the equations $a_2(x) = a$ and $C_2(x) = y$ force $3x + 1 = 2^a y$, hence $x = b_a(y)$.

Starting at x_* , a point of E_n is obtained by choosing a word $(a_1, \dots, a_n)$ of positive exponents and applying the corresponding inverse branches in the forced order. Thus E_n is the image of the countable set $\mathbb{N}_{>0}^n$, and E is a countable union of countable sets. If $x \in X_\infty$ but $C_2(x) \in E_n$, then $x \in E_{n+1}$, a contradiction. This proves forward invariance.

Putting $y = -1/3$ in the inverse-branch formula gives x_a . Moreover

$$3x_a + 1 = -\frac{2^a}{3},$$

whose valuation is a and whose quotient by 2^a is $-1/3$. □

2.2 Exact residue cylinders and survivor coding

For a finite word $w = (a_1, \dots, a_n)$, retain the notation

$$A(w) = a_1 + \dots + a_n, \quad B_w = \sum_{i=1}^n 3^{n-i} 2^{a_1 + \dots + a_{i-1}}$$

from Proposition 1.9; for the empty word, $A(\emptyset) = B_\emptyset = 0$. Define the odd 2-adic residue cylinder

$$\mathcal{R}_w = 3^{-n}(2^{A(w)} - B_w) + 2^{A(w)+1}\mathbb{Z}_2. \quad (35)$$

Here 3^{-n} is the inverse of 3^n in $\mathbb{Z}_2$.

Lemma 2.3 (exact 2-adic prefix cylinders). *For $x \in \mathcal{O}_2$, the following are equivalent:*

- (i) *the first n iterates of C_2 are defined at x and their successive valuation exponents are $a_1, \dots, a_n$;*
- (ii) *$x \in \mathcal{R}_w$;*
- (iii)

$$3^n x + B_w \equiv 2^{A(w)} \pmod{2^{A(w)+1}\mathbb{Z}_2}. \quad (36)$$

Every extension refines its prefix:

$$\mathcal{R}_{wa} \subseteq \mathcal{R}_w \quad (a \in \mathbb{N}_{>0}).$$

Proof. The equivalence of (ii) and (iii) follows by multiplication by the unit 3^n . We prove the equivalence with (i) by induction on n . The empty word gives $\mathcal{R}_\emptyset = 1 + 2\mathbb{Z}_2$. For $n = 1$, (36) is

$$3x + 1 \equiv 2^{a_1} \pmod{2^{a_1+1}\mathbb{Z}_2},$$

which is exactly $\nu_2(3x + 1) = a_1$.

For $n > 1$, put $w' = (a_2, \dots, a_n)$ and $A' = A(w) - a_1$. The identity

$$B_w = 3^{n-1} + 2^{a_1} B_{w'}$$

turns (36) into

$$3^{n-1}(3x + 1) + 2^{a_1} B_{w'} \equiv 2^{a_1+A'} \pmod{2^{a_1+A'+1}\mathbb{Z}_2}.$$

Reduction modulo 2^{a_1} first shows that 2^{a_1} divides $3x + 1$. After division by 2^{a_1} , the resulting congruence is

$$3^{n-1}y + B_{w'} \equiv 2^{A'} \pmod{2^{A'+1}\mathbb{Z}_2}.$$

where $y = (3x + 1)/2^{a_1}$. Its reduction modulo two says that y is odd, so the first valuation is exactly a_1 and $y = C_2(x)$. The induction hypothesis now proves that the remaining exponents are exactly w' . Every point with prefix wa also has prefix w , proving the final inclusion. This is the same integral congruence used for positive sources in Proposition 1.9; positivity is not used in the 2-adic argument. $\square$

Let

$$\Sigma = \mathbb{N}_{>0}^{\mathbb{N}_0}$$

with the product topology of the discrete alphabet, and let $\sigma(a_1, a_2, \dots) = (a_2, a_3, \dots)$. For a finite word $w = (a_1, \dots, a_n)$, write

$$[w] = \{\omega \in \Sigma : \omega_1 = a_1, \dots, \omega_n = a_n\}.$$

These finite cylinders form a basis of the product topology.

Theorem 2.4 (exact survivor itinerary coding). *The itinerary map*

$$\epsilon_2 : X_\infty \longrightarrow \Sigma, \quad \epsilon_2(x) = (a_2(x), a_2(C_2x), a_2(C_2^2x), \dots)$$

is a bijection. More precisely, if $\omega = (a_1, a_2, \dots)$ and $w_n = (a_1, \dots, a_n)$, then

$$\{\epsilon_2^{-1}(\omega)\} = \bigcap_{n \geq 0} \mathcal{R}_{w_n}. \quad (37)$$

For every finite word w , the cylinder identity is

$$\epsilon_2^{-1}([w]) = \mathcal{R}_w \cap X_\infty. \quad (38)$$

Consequently ϵ_2 is a homeomorphism.

Proof. For fixed ω , Lemma 2.3 gives a nested sequence of residue classes. Let r_n be any integral representative of $\mathcal{R}_{w_n}$. If $m \geq n$, then

$$r_m \equiv r_n \pmod{2^{A(w_n)+1}}.$$

Because $A(w_n) \geq n$, the sequence (r_n) is Cauchy in $\mathbb{Z}_2$. Completeness gives a limit $x(\omega)$, and the moduli tending to infinity make it unique. The limit lies in every $\mathcal{R}_{w_n}$.

The prefix lemma says that $x(\omega)$ has every finite prefix w_n . It cannot lie in E_m : if $C_2^m(x(\omega)) = x_*$, the exponent at position $m + 1$ would be undefined, contradicting membership in $\mathcal{R}_{w_{m+1}}$. Hence $x(\omega) \in X_\infty$ and $\epsilon_2(x(\omega)) = \omega$.

Conversely, every $x \in X_\infty$ has all iterates defined. Its first n exponents place it in the corresponding $\mathcal{R}_{w_n}$ for every n , and uniqueness of the intersection recovers x . This proves bijectivity and (37). The prefix lemma also proves (38). These sets are clopen in the relative 2-adic topology. Conversely, their moduli $2^{A(w_n)+1}$ tend to infinity, so the cylinders about any itinerary map inside arbitrarily small 2-adic neighborhoods. Thus both the map and its inverse are continuous. $\square$

Corollary 2.5 (shift conjugacy on the survivor space). *The self-map $C_2 : X_\infty \rightarrow X_\infty$ satisfies*

$$\epsilon_2 \circ C_2 = \sigma \circ \epsilon_2.$$

Proof. Forward invariance was proved in Proposition 2.2. Applying C_2 deletes the first valuation exponent and leaves every later exponent unchanged. $\square$

2.3 Positive itineraries and the canonical prefix tree

The positive odd set $\mathcal{O}$ of Section 1 is contained in X_∞ : its accelerated orbit remains positive and therefore never reaches $-1/3$. Define

$$\Sigma_C^+ = \epsilon_2(\mathcal{O}) \subset \Sigma.$$

Proposition 2.6 (exact position of the positive image). *The set Σ_C^+ is countable, dense in Σ , and forward-shift stable but not shift-surjective. More exactly,*

$$\sigma(\Sigma_C^+) = \epsilon_2(\{y \in \mathcal{O} : 3 \nmid y\}) \subsetneq \Sigma_C^+.$$

Every finite word occurs as the initial word of infinitely many positive odd orbits. These conclusions do not say that every infinite word belongs to Σ_C^+ .

Proof. Countability follows from countability of $\mathcal{O}$ and injectivity of ϵ_2 . Let w be a finite word. Its residue class $\mathcal{R}_w$ is odd modulo $2^{A(w)+1}$ and therefore contains infinitely many positive odd integers. By Lemma 2.3, each has prefix w . Thus every cylinder $[w]$ meets Σ_C^+ , proving density. Corollary 2.5 gives $\sigma(\Sigma_C^+) = \epsilon_2(C(\mathcal{O}))$. By Lemma 1.6, a positive odd integer belongs to $C(\mathcal{O})$ exactly when it is not divisible by three. The inclusion is strict because $3 \in \mathcal{O}$ has no positive odd predecessor, and injectivity of ϵ_2 on X_∞ prevents its itinerary from being the itinerary of another starting point. $\square$

Definition 2.7 (canonical prefix tree). For a finite word w , let $r(w)$ be the unique odd integer satisfying

$$1 \leq r(w) < 2^{A(w)+1}, \quad \mathcal{R}_w = r(w) + 2^{A(w)+1}\mathbb{Z}_2.$$

The canonical prefix tree has all finite positive-exponent words as vertices, an edge $w \rightarrow wa$ for every $a \in \mathbb{N}_{>0}$, and the typed vertex label

$$(A(w) + 1, r(w)).$$

Its level strictly increases along every edge.

The full word is retained in this tree. A proposed update of the residue label alone fails on a three-step orbit.

Proposition 2.8 (counterexample to the printed residue update). *The congruence rule*

$$3r' + 1 \equiv 2^a r \pmod{2^{N+a}} \tag{39}$$

does not update the canonical initial-source residue from a prefix label (N, r) to the label for its extension by a .

Indeed, the positive orbit beginning at 7 has exponent prefix $(1, 1, 2)$. The prefix $(1, 1)$ has $(N, r) = (3, 7)$. Substitution of $a = 2$ into (39) gives

$$3r' + 1 \equiv 28 \pmod{32}, \quad r' \equiv 9 \pmod{32},$$

whereas the exact cylinder formula gives

$$r(1, 1, 2) = 7 \pmod{32}.$$

Proof. The accelerated trajectory is

$$7 \xrightarrow{1} 11 \xrightarrow{1} 17 \xrightarrow{2} 13.$$

Thus its first two exponents give $A = 2$ and modulus $2^{A+1} = 8$, with canonical residue 7. The displayed update gives $3r' \equiv 27 \pmod{32}$ and hence $r' \equiv 9 \pmod{32}$.

For the full word, $A = 4$ and

$$B_{(1,1,2)} = 3^2 + 3 \cdot 2 + 2^2 = 19.$$

Equation (35) gives

$$r(1, 1, 2) \equiv 3^{-3}(16 - 19) \equiv 7 \pmod{32}.$$

The two residues differ, proving failure of the proposed rule. This counterexample concerns that formula; it does not by itself prove that no other update using additional typed data can be constructed. $\square$

For $\omega \in \Sigma$, write $w_n = \omega|n$ and $r_n(\omega) = r(w_n)$.

Theorem 2.9 (positive-integer stabilization criterion). *Let $x(\omega) = \epsilon_2^{-1}(\omega)$. Then*

$$x(\omega) \in \mathcal{O} \iff r_n(\omega) \text{ is eventually constant.}$$

When these conditions hold, the eventual constant is $x(\omega)$.

Proof. For every n ,

$$x(\omega) \equiv r_n(\omega) \pmod{2^{A(w_n)+1}\mathbb{Z}_2}.$$

If $x(\omega) = m$ is a positive odd integer, then for all sufficiently large n one has $2^{A(w_n)+1} > m$. The least positive representative of the residue class is then m , so $r_n(\omega) = m$.

Conversely, suppose $r_n(\omega) = m$ for every $n \geq n_0$. Then $x(\omega) - m$ is divisible in $\mathbb{Z}_2$ by $2^{A(w_n)+1}$ for arbitrarily large n . The only 2-adic integer with arbitrarily large valuation is zero, so $x(\omega) = m$. Every r_n is a positive odd integer, completing the proof. $\square$

Thus positive realizability is an exact asymptotic prefix-stabilization property of the entire rooted prefix chain. It is not invariant under finite changes of the initial symbols: $(2, 2, 2, \dots)$ codes the positive fixed point 1, whereas the tail-equivalent sequence $(1, 2, 2, \dots)$ codes $h_1(1) = 1/3$, which is not an integer. Density of Σ_C^+ shows that the property cannot be detected by excluding a finite list of finite words.

2.4 Elementary pressure on the full label shift

On the full shift Σ , define the one-coordinate potential

$$\phi(\omega) = \log 3 - a_1 \log 2 \quad (\omega = (a_1, a_2, \dots)).$$

For $\beta \in \mathbb{R}$ and $n \geq 1$, consider the nonnegative partition sum

$$Z_n(\beta) = \sum_{(a_1, \dots, a_n) \in \mathbb{N}_{>0}^n} \exp \left(\beta \sum_{j=1}^n (\log 3 - a_j \log 2) \right). \quad (40)$$

Proposition 2.10 (exact full-shift pressure and Bernoulli weights). *For $\beta > 0$, the partition sum converges and*

$$Z_n(\beta) = \left(\frac{(3/2)^\beta}{1 - 2^{-\beta}} \right)^n. \quad (41)$$

Hence its logarithmic pressure is

$$P(\beta) = \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_n(\beta) = \beta \log(3/2) - \log(1 - 2^{-\beta}). \quad (42)$$

The normalized one-symbol weights are

$$p_\beta(a) = (1 - 2^{-\beta})2^{-\beta(a-1)} \quad (a \geq 1). \quad (43)$$

At $\beta = 1$, $p_1(a) = 2^{-a}$. For $\beta \leq 0$, the one-symbol sum diverges and the finite formula above does not apply.

Proof. For $\beta > 0$, Tonelli's theorem for nonnegative series and independence of the coordinates give

$$Z_n(\beta) = \left(\sum_{a \geq 1} 3^\beta 2^{-\beta a} \right)^n.$$

The inner geometric series is

$$3^\beta \frac{2^{-\beta}}{1 - 2^{-\beta}} = \frac{(3/2)^\beta}{1 - 2^{-\beta}},$$

which proves (41); taking $n^{-1} \log$ proves (42). Dividing each one-symbol weight $3^\beta 2^{-\beta a}$ by their sum gives (43). If $\beta = 0$, every summand is one. If $\beta < 0$, the summands do not tend to zero as $a \rightarrow \infty$. Thus the one-symbol sum diverges for every $\beta \leq 0$. $\square$

The same calculation has an exact periodic-point formulation. This matters because the standard Gurevich pressure of a topologically mixing countable Markov shift is defined from periodic points returning to a fixed base cylinder, rather than from an unrestricted sum over all finite words [14, Sections 2 and 4].

Here Σ is the one-sided topological Markov shift of the complete directed graph on $\mathbb{N}_{>0}$, hence is topologically mixing. For every $\beta > 0$, the potential $\beta\phi$ is one-cylinder locally constant, has zero variations from level one onward, has summable variations, and satisfies $\sup \beta\phi = \beta \log(3/2) < \infty$. Formula (42) proves that its pressure is finite. Thus all hypotheses used to place the following periodic sum in Sarig's Gurevich-pressure framework are explicit rather than inferred from the phrase "full shift."

Proposition 2.11 (fixed-base periodic partition crosswalk). *Fix $b \in \mathbb{N}_{>0}$ and, for $\beta > 0$, put*

$$Z_n^{\text{per}}(\beta; b) = \sum_{\substack{\sigma^n \omega = \omega \\ \omega_1 = b}} \exp \left(\beta \sum_{j=0}^{n-1} \phi(\sigma^j \omega) \right).$$

Then

$$Z_n^{\text{per}}(\beta; b) = 3^\beta 2^{-\beta b} \left(\frac{(3/2)^\beta}{1 - 2^{-\beta}} \right)^{n-1}, \quad (44)$$

and therefore

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log Z_n^{\text{per}}(\beta; b) = P(\beta).$$

For $\beta \leq 0$, the fixed-base sum is infinite for every $n \geq 2$.

Proof. An n -periodic point with first symbol b is uniquely determined by the finite block $(b, a_2, \dots, a_n)$ repeated indefinitely. For $\beta > 0$, nonnegative factorization therefore gives

$$Z_n^{\text{per}}(\beta; b) = 3^\beta 2^{-\beta b} \left(\sum_{a \geq 1} 3^\beta 2^{-\beta a} \right)^{n-1}.$$

Substitution of the one-symbol geometric sum proves (44). The prefactor contributes $n^{-1}\beta(\log 3 - b \log 2)$ to the normalized logarithm, which tends to zero, leaving $P(\beta)$. If $\beta \leq 0$ and $n \geq 2$, the unrestricted second symbol already supplies a divergent one-symbol subseries. $\square$

Exact boundary 2.12. The preceding proposition is an elementary calculation on the full label shift, equivalently on X_∞ through Theorem 2.4. It does not turn the countable dense subset Σ_C^+ or the acyclic prefix-refinement tree into a closed recurrent Markov system. The periodic-point definition is crosswalked exactly, but no external recurrence, Gibbs, equilibrium, or transfer-operator theorem is invoked for either object.

2.5 The tropical degree and its exact typing boundary

The path bidegree

$$d(p) = (\ell(p), A(p) - \ell(p)) \in \mathbb{N}_0^2$$

is additive when paths are composable. That true statement does not make an object-erasing monomial assignment multiplicative on the ordinary category algebra.

Proposition 2.13 (exact fibres of the path bidegree). *The fibre of d over $(0, 0)$ consists of all identity paths. For every $s > 0$, the fibre over $(0, s)$ is empty. If $m \geq 1$ and $s \geq 0$, then the fibre over (m, s) is infinite and is partitioned by exactly*

$$\binom{m + s - 1}{m - 1}$$

exponent words, each having one infinite positive source class modulo 2^{m+s+1} .

Proof. A path of length zero is an identity and has total exponent zero, proving the first two statements. For length $m \geq 1$, the equation $d(p) = (m, s)$ is equivalent to

$$a_1 + \dots + a_m = m + s, \quad a_i \geq 1.$$

The number of such positive compositions is $\binom{m+s-1}{m-1}$. Proposition 1.9 assigns to each word one odd residue class modulo 2^{m+s+1} and infinitely many positive sources. Each source determines one typed path with that word, completing the fibre description. $\square$

Proposition 2.14 (object-idempotent obstruction to the monomial shadow). *Let R be a nonzero commutative ring, and let*

$$\eta : R[u, v] \longrightarrow R, \quad \eta(f) = f(1, 1).$$

The R -linear map

$$\pi_{\text{deg}} : \mathfrak{A}_C \longrightarrow R[X, Y], \quad \pi_{\text{deg}}(f(u, v)[p]) = \eta(f) X^{\ell(p)} Y^{A(p) - \ell(p)}$$

respects the product of two coefficient-decorated path basis terms whenever their paths are composable, but it is not an algebra homomorphism. More generally, no algebra homomorphism can send every path basis element $[p]$ to the displayed degree monomial while erasing every object idempotent.

Proof. The coefficient map η is a ring homomorphism. For composable p and q , both ℓ and A add, so $\pi_{\deg}(f[p]g[q]) = \pi_{\deg}(f[p])\pi_{\deg}(g[q])$ in the order fixed by the category-algebra convention. Choose distinct objects $x, y \in \mathcal{O}$. In the category algebra,

$$e_x e_y = 0.$$

Both empty paths have degree $(0, 0)$, so the proposed assignment sends each of e_x, e_y to 1. Their images multiply to $1 \neq 0$, contradicting multiplicativity. Proposition 1.23 gives the exact repair: object matrix units retain orthogonality and make noncomposable products zero in the target. $\square$

Source repair 2.15 (arithmetic-site comparison through principal upper quadrants). Connes and Consani construct a semiring $\text{Conv}(\mathbb{Z} \times \mathbb{Z})$ of closed convex upper subsets with integral extreme points and identify it with a stated quotient of the unreduced square of the arithmetic site [11, source TeX lines 1151–1214 and 1364–1410]. Their coordinate Frobenius operations belong to that sourced construction; the source also distinguishes diagonal Frobenius from the ordinary power map and records noncancellativity of the unreduced tensor.

Put $Q = \mathbb{R}_{\geq 0}^2$. The source’s multiplicative unit is the upper quadrant Q , not the singleton $\{(0, 0)\}$. The exact principal-polygon map from the Collatz degree monoid is

$$j: \mathbb{N}_0^2 \longrightarrow \text{Conv}(\mathbb{Z} \times \mathbb{Z}), \quad j(r, s) = (r, s) + Q. \quad (45)$$

For every $(r, s) \in \mathbb{N}_0^2$, the set $(r, s) + Q$ is closed and convex, is Q -upper because $Q + Q = Q$, is contained in the translate $(r, s) + Q$, and has the unique integral extreme point (r, s) . It is therefore an element of the source’s $\text{Conv}(\mathbb{Z} \times \mathbb{Z})$. Its unique extreme point also proves that j is injective. Moreover $j(0, 0) = Q$, and Minkowski multiplication gives

$$j(r, s) j(r', s') = j(r + r', s + s').$$

Thus $j \circ d$ respects every composable path product. It also retains the empty degree fibres $(0, s)$ for $s > 0$ proved in Proposition 2.13; no singleton polygon is inserted for them.

To state the zero obstruction without inventing a coefficient map, let $\text{Mor}(\mathcal{E}_C)^0$ be the path semigroup with an absorbing zero: composable paths multiply by category composition and noncomposable paths multiply to 0. The only zero-preserving extension of the displayed object-erasing rule is

$$\theta(0) = \emptyset, \quad \theta(p) = j(d(p)).$$

It is multiplicative on every composable pair. It is not a semigroup homomorphism: for distinct objects $x \neq y$ one has $e_x e_y = 0$, whereas

$$\theta(e_x)\theta(e_y) = Q +_{\text{Mink}} Q = Q \neq \emptyset = \theta(0).$$

Thus j is a valid monoid embedding of degrees, while θ is not a multiplicative path-semigroup-with-zero representation. No algebra map to the convex semiring has been defined because no coefficient-semiring map has been supplied. This is a proved boundary for the displayed assignment, not a claim that no future typed morphism can exist.

2.6 Formal-scheme and prismatic typing boundaries

For each $a \geq 1$, the formula

$$g_a(x) = \frac{3x + 1}{2^a}$$

is a continuous affine set map $\mathbb{Z}_3 \rightarrow \mathbb{Z}_3$, since 2 is a unit in $\mathbb{Z}_3$. A self-map of the underlying set is not, by itself, a morphism of the formal scheme $\mathrm{Spf}(\mathbb{Z}_3)$.

Theorem 2.16 (no Collatz branch endomorphism of $\mathrm{Spf}(\mathbb{Z}_3)$). *Every continuous unital ring endomorphism $h: \mathbb{Z}_3 \rightarrow \mathbb{Z}_3$ is the identity. Consequently no affine branch g_a is induced by a formal-scheme endomorphism*

$$\mathrm{Spf}(\mathbb{Z}_3) \longrightarrow \mathrm{Spf}(\mathbb{Z}_3).$$

Proof. A unital ring homomorphism fixes the image of $\mathbb{Z}$ pointwise. The ordinary integers are dense in $\mathbb{Z}_3$, so continuity forces h to fix all of $\mathbb{Z}_3$. On the other hand,

$$g_a(0) = 2^{-a} \neq 0.$$

Thus g_a is not a ring endomorphism and cannot induce the displayed formal scheme morphism. $\square$

The fact that $(2^m) = \mathbb{Z}_3$ for every m proves that 2-power ideals do not provide a filtration on the formal point. The stronger statement needed for the integer exponent selector is the following exact Chinese-remainder theorem.

Proposition 2.17 (the integer valuation strata are 3-adically dense and codense). *For $a \geq 1$, put*

$$\mathcal{D}_a^+ = \{n \in \mathcal{O} : \nu_2(3n + 1) = a\}.$$

For every $k \geq 1$ and every residue $b \pmod{3^k}$, there is a positive odd $n \equiv b \pmod{3^k}$ in $\mathcal{D}_a^+$, and there is a positive odd $m \equiv b \pmod{3^k}$ outside $\mathcal{D}_a^+$; the latter may be chosen in $\mathcal{D}_{a+1}^+$. Consequently both $\mathcal{D}_a^+$ and its complement inside $\mathcal{O}$ meet every 3-adic residue ball. In particular, no clopen subset $U \subset \mathbb{Z}_3$ satisfies $U \cap \mathcal{O} = \mathcal{D}_a^+$.

Proof. Because 3 is invertible modulo 2^{a+1} , let

$$c_a \equiv 3^{-1}(2^a - 1) \pmod{2^{a+1}}.$$

This residue is odd and satisfies $3c_a + 1 \equiv 2^a \pmod{2^{a+1}}$. The moduli 3^k and 2^{a+1} are coprime, so the Chinese remainder theorem gives an integer satisfying

$$n \equiv b \pmod{3^k}, \quad n \equiv c_a \pmod{2^{a+1}}.$$

Adding a sufficiently large positive multiple of $3^k 2^{a+1}$ makes n positive without changing either congruence. It remains odd, and $3n + 1 \equiv 2^a \pmod{2^{a+1}}$, so $\nu_2(3n + 1) = a$. Repeating the construction with c_{a+1} modulo 2^{a+2} gives a positive odd m in the same 3-adic residue ball with valuation $a + 1$, hence outside $\mathcal{D}_a^+$.

If a clopen U cut out $\mathcal{D}_a^+$ on $\mathcal{O}$, choose $n \in \mathcal{D}_a^+ \subset U$. Openness supplies a residue ball $n + 3^k \mathbb{Z}_3 \subset U$. The preceding construction puts a positive odd $m \notin \mathcal{D}_a^+$ in that same ball, contradicting $U \cap \mathcal{O} = \mathcal{D}_a^+$. $\square$

Source repair 2.18 (Cartier–Witt functoriality and cyclotomic scope). Bhatt and Lurie define the Cartier–Witt stack WCart_X for a bounded p -adic formal scheme X and obtain functoriality from an actual formal-scheme map $f : X \rightarrow Y$ [12, source TeX lines 480–550]. Their Witt-vector Frobenius is a separate endomorphism in that construction. By Theorem 2.16, their functoriality does not produce a map denoted $\mathrm{WCart}(g_a)$ on $X = \mathrm{Spf}(\mathbb{Z}_3)$ from the affine Collatz formula.

Gros, Le Stum, and Quirós prove one-variable cyclotomic equivalences: Theorem 7.5 uses complete modules with Δ -hyperstratification, while Theorem 7.12 identifies the finite-free weakly nilpotent ∇_Δ -module and vector-bundle categories [13, Theorems 7.5 and 7.12]. Their hypotheses use a Witt ring W , a primitive p th root of unity, and the one-variable ring $W[[q - 1]]$. Those theorems do not supply two independent Collatz coordinate directions, two commuting connection operators, or compatibility with the path category. Constructing such an object would require a new base, modules, operators, and comparison maps with their hypotheses proved.

Exact boundary 2.19. The chapter constructs no tropical Collatz semiring, characteristic-one functor, Cartier–Witt action of the affine branches, prismatic lift of the path category, groupoid completion, or operator-algebra completion. The two obstructions prove that the displayed object-erasing algebra assignment and the proposed formal base do not work as stated. They do not assert that every better-typed interface is impossible. None of the symbolic or interface results proves termination of every positive Collatz orbit.

The portable certificate `certificates/finite_actions_checks.py` checks the first exceptional preimages for $1 \leq a \leq 24$; prefix refinements for parent-word lengths 0, 1, 2, 3 with parent and child symbols in $\{1, 2, 3, 4\}$; the explicit failed residue update; finite-alphabet pressure factorizations for $1 \leq \beta \leq 4$, alphabet bounds $1 \leq B \leq 6$, and word lengths $1 \leq n \leq 3$; and bounded CRT selector instances recorded by the script. These checks support exactly those finite ranges. The infinite survivor bijection, homeomorphism, stabilization theorem, CRT density theorem, and source-typing proofs are established by the displayed arguments rather than by bounded enumeration.

3 Arithmetic orbit groupoids and toric period morphisms

This section constructs two interfaces which begin with the same positive-odd path coordinates but retain different information. The first is the discrete orbit groupoid of the actual positive-integer map. Its singletons are compact open bisections, so the finite-support path algebra embeds by point masses. The second is the germ groupoid generated by exact partial homeomorphisms of $\mathbb{Z}_3$. Its unit space is compact and its arrows vary 3-adically; arithmetic paths enter it by a faithful functor, not by identifying its convolution algebra with the discrete path algebra.

The period interface is equally explicit. A unimodular change of torus coordinates identifies the path bidegree $(\ell, A - \ell)$ with the character $(A, -\ell)$, and every logarithm used below is obtained from a commutative square of Deligne 1-motives. The arctangent integral is related to a Kummer logarithm by an isomorphism of algebraic pairs. These are actual maps; no categorical relation is inferred from similar formulas.

Deaconu’s construction is the literature prototype for a groupoid attached to an endomorphism, but his topological theorem assumes a compact Hausdorff space and a self-covering map [18, printed pp. 1779–1781]. Our discrete theorem is therefore proved directly. The definitions and period computations for 1-motives are crosswalked against Huber–Wüstholz [19, Definitions 8.1 and 10.2; Proposition 12.10]; the permitted arrows in Nori’s diagram of pairs are those of Huber–Müller-Stach [20, Definitions 7.1.4 and 9.1.1]. Barbieri-Viale–Kahn supply the general derived

and Cartier-dual framework [21, Section 1.1 and Theorem 2.1.2]. None of these sources states the Collatz-specific constructions proved here.

3.1 The discrete arithmetic orbit groupoid

Retain the positive-odd map C , exponent function a , and first-return category $\mathcal{E}_C$ from Definition 1.1. For $n \geq 0$ put

$$S_n a(x) = \sum_{j=0}^{n-1} a(C^j x), \quad S_0 a(x) = 0.$$

Definition 3.1 (arithmetic orbit groupoid). Let

$$\mathcal{G}_{\text{arith}} = \left\{ (x, k, y) \in \mathcal{O} \times \mathbb{Z} \times \mathcal{O} : \begin{array}{l} \text{there are } m, n \geq 0 \text{ with } k = m - n, \\ C^m(x) = C^n(y) \end{array} \right\}. \quad (46)$$

Its range and source are $r(x, k, y) = x$ and $s(x, k, y) = y$, and

$$(x, k, y)(y, l, z) = (x, k + l, z), \quad (x, k, y)^{-1} = (y, -k, x).$$

We give $\mathcal{G}_{\text{arith}}$ the discrete topology.

Theorem 3.2 (exact arithmetic groupoid and its two cocycles). *The preceding formulas define a countable, Hausdorff, locally compact, étale groupoid. Counting measure on every range fibre is a Haar system. For $g = (x, k, y)$ choose any $m, n \geq 0$ representing it and put*

$$c_a(g) = S_m a(x) - S_n a(y). \quad (47)$$

This integer does not depend on the representative. The maps

$$\kappa(g) = k, \quad (48)$$

$$D(g) = (-k, -c_a(g) + k) \in \mathbb{Z}^2, \quad (49)$$

$$\delta_G(g) = -c_a(g) \log 2 + k \log 3 \quad (50)$$

are groupoid homomorphisms to the displayed additive groups.

Proof. If (m, n) represents (x, k, y) and $t \geq 0$, then $(m + t, n + t)$ represents the same triple. With $u = C^m(x) = C^n(y)$, the Birkhoff-sum identity gives

$$S_{m+t} a(x) = S_m a(x) + S_t a(u), \quad S_{n+t} a(y) = S_n a(y) + S_t a(u),$$

so their difference is unchanged. Any two representatives have the same difference $m - n$; after extending the shorter pair, both reach a common pair of times. This proves that (47) is well-defined.

For multiplication, suppose (m, n) represents $g = (x, k, y)$ and (p, q) represents $h = (y, l, z)$. Put $L = \max\{n, p\}$. Extending the first pair by $L - n$ and the second by $L - p$ gives a representative

$$(m + L - n, q + L - p)$$

of gh . The two sums along the common middle orbit cancel, giving $c_a(gh) = c_a(g) + c_a(h)$. The same aligned representative gives $\kappa(gh) = \kappa(g) + \kappa(h)$. Equations (49) and (50) are integral and real linear combinations of these two cocycles, so they are additive as well.

The groupoid axioms now follow from the displayed product and inverse. The underlying set is countable. In the discrete topology every singleton is a compact open bisection, range and source are local homeomorphisms, and compact subsets are finite. Thus the groupoid is Hausdorff, locally compact, and étale. Left multiplication bijects range fibres, so counting measures are left invariant and form the counting Haar system. $\square$

Theorem 3.3 (faithful path embedding and operator actions). *For a path $p : x \rightarrow y$ define*

$$J(p) = (y, -\ell(p), x). \quad (51)$$

Then $J : \mathcal{E}_C \rightarrow \mathcal{G}_{\text{arith}}$ is an injective functor, where the target is regarded as a groupoid category. On its image,

$$D(J(p)) = (\ell(p), A(p) - \ell(p)), \quad \delta_{\mathcal{G}}(J(p)) = A(p) \log 2 - \ell(p) \log 3. \quad (52)$$

Consequently the point-mass map

$$\mathbb{C}[\mathcal{E}_C] \longrightarrow C_c(\mathcal{G}_{\text{arith}}), \quad [p] \longmapsto 1_{\{J(p)\}},$$

is an injective algebra homomorphism in the target-left convention.

For $z = (z_1, z_2) \in \mathbb{T}^2$ and $t \in \mathbb{R}$, the formulas

$$(\Gamma_z f)(g) = z_1^{D_1(g)} z_2^{D_2(g)} f(g), \quad (53)$$

$$(\alpha_t f)(g) = e^{it\delta_{\mathcal{G}}(g)} f(g) \quad (54)$$

define commuting automorphism groups of the convolution algebra. They extend separately to both the full and reduced groupoid C^ -algebras.*

Proof. The pair $(0, \ell(p))$ represents $J(p)$ because $C^0(y) = C^{\ell(p)}(x)$. If $p : x \rightarrow y$ and $q : z \rightarrow x$, then

$$J(p)J(q) = (y, -\ell(p) - \ell(q), z) = J(p \circ q),$$

which is exactly the target-left multiplication used in (1). A path of the deterministic category is fixed by its source and length, so J is injective. The same representative has $c_a(J(p)) = -A(p)$; substitution gives (52). Distinct point masses are linearly independent, proving the algebra assertion.

Additivity of D and $\delta_{\mathcal{G}}$ makes (53) and (54) multiplicative under convolution and compatible with involution. Their inverses are obtained from z^{-1} and $-t$. The universal property gives the full extensions. In every regular representation the same formulas are implemented by diagonal unitaries, so they preserve the reduced norm and give the reduced extensions. $\square$

Exact boundary 3.4. No amenability assertion or equality between the full and reduced C^* -algebras is used. Equations (53)–(54) construct dynamics; they do not classify equilibrium states or prove a phase transition.

3.2 The compact 3-adic pseudogroup and the arithmetic functor

Set $X = \mathbb{Z}_3$. For $a \geq 1$ put

$$Y_a = \{y \in X : 2^a y \equiv 1 \pmod{3}\}, \quad g_a(x) = \frac{3x+1}{2^a}, \quad \beta_a(y) = \frac{2^a y - 1}{3}. \quad (55)$$

Proposition 3.5 (exact 3-adic branch domains). *The set Y_a is the clopen residue class $2^{-a} + 3\mathbb{Z}_3$. The maps*

$$g_a : X \longrightarrow Y_a, \quad \beta_a : Y_a \longrightarrow X$$

are mutually inverse homeomorphisms. For a chronological word $w = (a_1, \dots, a_n)$, the composite

$$h_w = g_{a_n} \circ \dots \circ g_{a_1} : X \longrightarrow Y_w := h_w(X)$$

is a homeomorphism onto a clopen set and has the affine formula

$$h_w(x) = \frac{3^n x + B_w}{2^{A(w)}}, \quad B_w = \sum_{i=1}^n 3^{n-i} 2^{a_1 + \dots + a_{i-1}}. \quad (56)$$

Proof. The congruence defining Y_a is one residue class because 2^a is a unit in $\mathbb{Z}_3$. The numerator of β_a lies in $3\mathbb{Z}_3$ on that class, and direct substitution gives

$$\beta_a g_a(x) = x, \quad g_a \beta_a(y) = y.$$

Both maps are continuous affine maps, hence are inverse homeomorphisms. Composition proves the topological assertion for words; induction gives (56). $\square$

Let $\mathcal{P}_3$ be the pseudogroup generated by restrictions of the g_a and β_a , and let $\mathcal{H}_3$ be its groupoid of germs. We write $[\phi, x]$ for the germ at x of a pseudogroup element ϕ .

Theorem 3.6 (the 3-adic germ groupoid and its arithmetic subgroupoid). *The groupoid $\mathcal{H}_3$ is second countable, Hausdorff, locally compact, and étale, with compact unit space X and its counting Haar system.*

For $g = (x, m - n, y) \in \mathcal{G}_{\text{arith}}$, let $p_m(x)$ and $p_n(y)$ be the actual chronological exponent words of the first m and n steps. Then

$$\Xi(g) = [h_{p_m(x)}^{-1} \circ h_{p_n(y)}, y] \quad (57)$$

is a well-defined injective groupoid functor, with object map the inclusion $\mathcal{O} \hookrightarrow \mathbb{Z}_3$, $\Xi : \mathcal{G}_{\text{arith}} \rightarrow \mathcal{H}_3$. It sends $J(p)$ to $[h_p, s(p)]$.

If $u(g)$ is the rational slope of an affine representative of $\Xi(g)$, then

$$u(g) = 3^{n-m} 2^{S_m a(x) - S_n a(y)}, \quad D(g) = (\nu_3(u(g)), -\nu_2(u(g)) - \nu_3(u(g))), \quad (58)$$

and

$$\delta_{\mathcal{G}}(g) = -\log |u(g)|_{\infty}. \quad (59)$$

Thus Ξ preserves both the bidegree and real cocycle.

Proof. Every element of $\mathcal{P}_3$ is locally the restriction of an affine map $x \mapsto ux + v$ with $u \neq 0$. Two such maps agreeing on a nonempty 3-adic open set have the same slope and intercept. Hence two distinct germs with a common source can be separated by basic germ bisections; germs with different sources are separated in the unit space. This proves Hausdorffness. The usual germ bisections make range and source local homeomorphisms. Since $\mathbb{Z}_3$ has a countable clopen basis and the generating words are countable, the groupoid is second countable. Every germ has a bisection with compact-open source, proving local compactness. Counting measures are the canonical Haar system of an étale groupoid.

The equality $C^m(x) = C^n(y)$ says that the two word maps in (57) meet at their displayed points, so the composite germ is defined near y and maps it to x . Replacing (m, n) by $(m+t, n+t)$ prefixes both word maps by the same future affine map h_r ; the factor h_r cancels between inverse and forward maps. Two arbitrary representatives admit a common extension, proving well-definedness. The same common-middle alignment used in Theorem 3.2 proves functoriality.

The slope of $h_{p_m(x)}$ is $3^m 2^{-S_m a(x)}$, so the slope of its inverse followed by $h_{p_n(y)}$ is the first expression in (58). Its 3-adic valuation is $n - m$, which recovers the integer coordinate $m - n$. Source, range, and this coordinate recover the entire triple $(x, m - n, y)$, so Ξ is injective. The two valuation formulas and (59) follow by substitution. For $J(p)$, the representative is $(0, \ell(p))$, so its germ is h_p . $\square$

The functor Ξ is not an inclusion of convolution algebras by point masses: a singleton germ over the nondiscrete unit space $\mathbb{Z}_3$ is generally not open, so its characteristic function is not in $C_c(\mathcal{H}_3)$. It is also not legitimate to obtain an operator model on $\ell^2(\mathcal{O})$ by restricting every g_a .

Proposition 3.7 (explicit failure of the unrestricted arithmetic restriction). *The 3-adic homeomorphism $g_a : X \rightarrow Y_a$ does not preserve $\mathcal{O}$. In particular $g_1(1) = 2$. If one defines*

$$(V_a \xi)(y) = \begin{cases} \xi(\beta_a(y)), & y \in Y_a \cap \mathcal{O}, \\ 0, & y \notin Y_a \cap \mathcal{O}, \end{cases}$$

on $\ell^2(\mathcal{O})$, then $V_1 \delta_1 = 0$ and hence $V_1^* V_1 \neq 1$.

Proof. The equation $\beta_1(y) = 1$ has the unique solution $y = 2$, which is not a positive odd integer. Therefore every coordinate of $V_1 \delta_1$ vanishes. $\square$

There is a second elementary obstruction to a commonly proposed operator symmetry. For a nonempty free word q , left prefixing $L_q[p] = [qp]$ is not an algebra endomorphism of the free-monoid algebra, since

$$L_q(1 \cdot 1) = [q] \neq [qq] = L_q(1)L_q(1).$$

Consequently no prefix endomorphism of a completion follows from word concatenation alone; its action on the diagonal and all covariance relations would have to be declared and proved separately.

Exact boundary 3.8. The exact objects constructed here are $\mathcal{G}_{\text{arith}}$, $\mathcal{H}_3$, and the faithful functor Ξ . No unspecified semigroup crossed product is identified with either full or reduced groupoid C^* -algebra. Such an identification requires a declared covariance relation and a proof of the corresponding universal property.

3.3 The exact torus coordinate change

Work over $\mathbb{Q}$. Use two copies of the split torus, with coordinates

$$T_{\text{ar}} = (\mathbf{G}_m)_{x,y}^2, \quad T_{\text{deg}} = (\mathbf{G}_m)_{\lambda,\mu}^2.$$

For a path p put

$$\chi_p(x, y) = x^{A(p)} y^{-\ell(p)}, \quad \psi_p(\lambda, \mu) = \lambda^{\ell(p)} \mu^{A(p) - \ell(p)}.$$

Theorem 3.9 (unimodular arithmetic-degree torus isomorphism). *The maps*

$$\Phi : T_{\text{ar}} \longrightarrow T_{\text{deg}}, \quad \Phi(x, y) = (x/y, x), \quad \Phi^{-1}(\lambda, \mu) = (\mu, \mu/\lambda) \quad (60)$$

are inverse torus isomorphisms, and

$$\psi_p \circ \Phi = \chi_p. \quad (61)$$

On exponent lattices the mutually inverse matrices are

$$\begin{pmatrix} A \\ -\ell \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \ell \\ A - \ell \end{pmatrix}, \quad \begin{pmatrix} \ell \\ A - \ell \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} A \\ -\ell \end{pmatrix}. \quad (62)$$

Proof. Direct substitution proves the inverse formulas and

$$\psi_p(\Phi(x, y)) = (x/y)^{\ell(p)} x^{A(p)-\ell(p)} = x^{A(p)} y^{-\ell(p)}.$$

Multiplication of the two integral matrices gives the identity in both orders, proving the lattice statement without changing either coordinate system. $\square$

Define the two isomorphic 1-motives

$$M_{\text{ar}} = [\mathbb{Z} \xrightarrow{u_{\text{ar}}} T_{\text{ar}}], \quad u_{\text{ar}}(n) = (2^n, 3^n), \quad (63)$$

$$M_{\text{deg}} = [\mathbb{Z} \xrightarrow{u_{\text{deg}}} T_{\text{deg}}], \quad u_{\text{deg}}(n) = ((2/3)^n, 2^n). \quad (64)$$

Then $(\text{id}_{\mathbb{Z}}, \Phi)$ is an isomorphism $M_{\text{ar}} \rightarrow M_{\text{deg}}$.

Theorem 3.10 (character-induced Kummer morphisms and exact periods). *For a path p set*

$$\alpha_p = 2^{A(p)} 3^{-\ell(p)}, \quad K(\alpha_p) = [\mathbb{Z} \xrightarrow{n \mapsto \alpha_p^n} \mathbf{G}_m].$$

Then

$$(\text{id}_{\mathbb{Z}}, \chi_p) : M_{\text{ar}} \longrightarrow K(\alpha_p) \quad (65)$$

is a morphism of 1-motives. Along the positive-real path $\gamma(s) = (2^s, 3^s)$, $0 \leq s \leq 1$,

$$\int_{\gamma} \chi_p^* \frac{dz}{z} = A(p) \log 2 - \ell(p) \log 3. \quad (66)$$

The branch is fixed by this path.

For every prefix $1 \leq k \leq \ell(p)$, the character

$$\chi_{p,k}(x, y) = x^{-A_k(p)} y^{k-1}$$

similarly gives a morphism to $K(r_k(p))$, where $r_k(p) = 3^{k-1} 2^{-A_k(p)}$, and its chosen period is $\log r_k(p)$.

Proof. A morphism $[L \rightarrow G] \rightarrow [L' \rightarrow G']$ of 1-motives is a commutative square [19, Definition 8.1]. Here

$$\chi_p(u_{\text{ar}}(n)) = (2^n)^{A(p)} (3^n)^{-\ell(p)} = \alpha_p^n,$$

so (65) is such a square. Moreover

$$\chi_p^* \frac{dz}{z} = A(p) \frac{dx}{x} - \ell(p) \frac{dy}{y}.$$

Integration on γ proves (66). The prefix statement is the same calculation with exponent vector $(-A_k(p), k-1)$. $\square$

Cartier duality makes the fixed rank-two object equally explicit. The dual of M_{ar} has the form

$$M_{\text{ar}}^\vee = [\mathbb{Z}^2 \longrightarrow \mathbf{G}_m], \quad (r, s) \longmapsto 2^r 3^s. \quad (67)$$

Thus $(A(p), -\ell(p))$ evaluates to α_p . Formula (67) is the specialization of the character lattice in Cartier duality, not a second unidentified motive [21, Section 1.8].

3.4 Cayley pairs and the exact Nori arrows

Let $K = \mathbb{Q}(i)$ and

$$U_K = \mathbf{P}_K^1 \setminus \{i, -i\}.$$

Theorem 3.11 (Cayley isomorphism of pairs). *The rational maps*

$$\kappa(z) = \frac{1+iz}{1-iz}, \quad \kappa^{-1}(w) = \frac{w-1}{i(w+1)} \quad (68)$$

are inverse isomorphisms $U_K \simeq \mathbf{G}_{m,K}$. For $r \in \mathbb{Q}_{>0}$ and $\alpha(r) = (1+ir)/(1-ir)$ they give an isomorphism of pairs

$$\kappa : (U_K, \{0, r\}) \xrightarrow{\sim} (\mathbf{G}_{m,K}, \{1, \alpha(r)\}), \quad (69)$$

and

$$\kappa^* \frac{dw}{w} = \frac{2i dz}{1+z^2}. \quad (70)$$

Consequently the functoriality edge in Nori's effective diagram has the contravariant direction

$$(\mathbf{G}_{m,K}, \{1, \alpha(r)\}, 1) \longrightarrow (U_K, \{0, r\}, 1). \quad (71)$$

Proof. Solving $w = (1+iz)/(1-iz)$ gives the displayed inverse. The two deleted points map to 0 and ∞ , so the restriction is an isomorphism to $\mathbf{G}_m$. It sends 0 to 1 and r to $\alpha(r)$, proving the pair statement. Logarithmic differentiation gives

$$\frac{i dz}{1+iz} + \frac{i dz}{1-iz} = \frac{2i dz}{1+z^2}.$$

Definition 9.1.1 of [20] sends a map of pairs contravariantly on relative cohomology, which fixes the edge direction. $\square$

Over $\mathbb{Q}$ itself, put

$$U_{\mathbb{Q}} = \mathbf{P}_{\mathbb{Q}}^1 \setminus V(z^2 + 1).$$

The form $dz/(1+z^2)$ is regular on $U_{\mathbb{Q}}$, including at infinity, and $0, r$ are rational points. Hence

$$\int_0^r \frac{dz}{1+z^2} = \arctan r$$

is a 1-period of the pair $(U_{\mathbb{Q}}, \{0, r\}, 1)$. Proposition 12.10 of [19] identifies this with the period world of 1-motives. More explicitly, with the local coordinate $t = 1/z$ at infinity,

$$\frac{dz}{1+z^2} = -\frac{dt}{1+t^2},$$

which proves regularity there. Removing infinity is therefore unnecessary, and doing so changes the Cayley target to $\mathbf{G}_m \setminus \{-1\}$.

Path concatenation by itself is not a Nori edge. A Nori functoriality edge requires a map of algebraic pairs, and a coboundary edge requires a nested triple of closed subvarieties. The exact replacement for additive path cocycles uses multiplication.

Proposition 3.12 (multiplication diagram for concatenated characters). *For composable paths p, q ,*

$$\chi_{q \circ p} = \chi_q \chi_p, \quad \alpha_{q \circ p} = \alpha_q \alpha_p.$$

Let

$$N_{p,q} = [\mathbb{Z} \longrightarrow \mathbf{G}_m^2], \quad n \longmapsto (\alpha_p^n, \alpha_q^n).$$

There is a factorization by morphisms of 1-motives

$$M_{\text{ar}} \xrightarrow{(\text{id}, (\chi_p, \chi_q))} N_{p,q} \xrightarrow{(\text{id}, m)} K(\alpha_p \alpha_q), \quad m(z_1, z_2) = z_1 z_2. \quad (72)$$

At the level of Nori pairs, the map

$$m : (\mathbf{G}_m^2, \{(1, 1), (\alpha_p, \alpha_q)\}) \longrightarrow (\mathbf{G}_m, \{1, \alpha_p \alpha_q\})$$

gives the contravariant functoriality edge

$$(\mathbf{G}_m, \{1, \alpha_p \alpha_q\}, 1) \xrightarrow{m^*} (\mathbf{G}_m^2, \{(1, 1), (\alpha_p, \alpha_q)\}, 1),$$

and

$$m^* \frac{dz}{z} = \text{pr}_1^* \frac{dz}{z} + \text{pr}_2^* \frac{dz}{z}. \quad (73)$$

Thus this declared pair map, rather than concatenation of singular paths, is the exact diagrammatic source of period additivity.

Proof. Lengths and total exponents add under composition, which proves the two character identities. Both squares in (72) commute on $n \in \mathbb{Z}$: the first maps $u_{\text{ar}}(n)$ to (α_p^n, α_q^n) , and multiplication maps that point to $(\alpha_p \alpha_q)^n$. The displayed map sends the two marked points into the two marked points, so it is a map of pairs. Logarithmic differentiation proves (73). $\square$

For $r > 0$,

$$\text{Im } \alpha(r) = \frac{2r}{1+r^2} > 0.$$

It therefore cannot equal the positive rational number $2^{A(p)}3^{-\ell(p)}$. The Cayley pair for a prefix angle and the rational Kummer pair for the path weight are linked by the toric constructions above, but they are not identified by an unproved equality of marked points.

Exact boundary 3.13. The section constructs the arithmetic groupoid, the germ groupoid with compact unit space, their faithful cocycle-preserving functor, the torus isomorphism, the Kummer morphisms, the Cayley pair isomorphism, and the multiplication edges. It does not construct a universal path-indexed Nori category, a classification of equilibrium states, a motivic Galois action on the path algebra, or a Collatz termination theorem.

4 Affine packets, parity necklaces, and exact residue automata

This section reconstructs an exponent-packet programme that arose in the modern Chatnotes corpus. The fixed-return formula itself is classical: Böhm–Sontacchi and Crandall give parity-coordinate forms, Lagarias proves the rational parity-word correspondence, and Urata writes the odd-step exponent formula and asks when its rational periodic points are integral [1, 2, 3, 5]. The comparison below identifies the rational points, transports cyclic rotation together with its stabilizers, and relates the full, shortened, and odd-return clocks. It also resolves the packet label (m, A) into an exact fixed-content primitive-necklace count and a packetwise sign split. The same exponent coordinates give a finite-state recursion for numerator residues, retaining their multiplicities.

4.1 Odd-step coordinates and the classical parity numerator

Let $\mathbb{Z}_{(2)} = \{u/v \in \mathbb{Q} : u, v \in \mathbb{Z}, v \text{ odd}\}$. The parity of u/v is $u \bmod 2$; it is independent of the odd-denominator representation. Both maps

$$T(x) = \begin{cases} (3x+1)/2, & x \text{ odd}, \\ x/2, & x \text{ even}, \end{cases} \quad F(x) = \begin{cases} 3x+1, & x \text{ odd}, \\ x/2, & x \text{ even} \end{cases}$$

take $\mathbb{Z}_{(2)}$ to itself. We call T the shortened map and F the full map. Lagarias uses T in his rational-cycle theorem [3, (1.1), Theorem 2.1]. Let

$$\mathcal{U}_2 = \left\{ \frac{u}{v} \in \mathbb{Q} : u, v \in \mathbb{Z} \text{ odd}, v \neq 0 \right\}.$$

For $x \in \mathcal{U}_2 \setminus \{-1/3\}$, the numerator of $3x+1$ in odd-denominator form is even. Thus

$$\Phi(x) = \frac{3x+1}{2^{\nu_2(3x+1)}} \quad \text{and} \quad a(x) = \nu_2(3x+1) \in \mathbb{N}_{>0}$$

are well defined and $\Phi(x) \in \mathcal{U}_2$.

For $m \geq 1$ and a packet $p = (a_1, \dots, a_m) \in \mathbb{N}_{>0}^m$ put

$$A = A(p) = \sum_{i=1}^m a_i, \quad S_0 = 0, \quad S_j = \sum_{i=1}^j a_i,$$

and define

$$C(p) = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{S_j}, \quad D_{m,A} = 2^A - 3^m, \quad x_p = \frac{C(p)}{D_{m,A}}. \quad (74)$$

The equality $2^A = 3^m$ is impossible, so $D_{m,A} \neq 0$. Moreover $C(p)$ and $D_{m,A}$ are odd. For tail recurrences only, set $A(\emptyset) = C(\emptyset) = 0$; no point $x_\emptyset$ is defined.

Write

$$\Pi_{m,A} = \{p \in \mathbb{N}_{>0}^m : A(p) = A\}.$$

Here A is a nonnegative integer, and $\Pi_{m,A} = \emptyset$ when $A < m$. For $A \geq m$ and $p \in \Pi_{m,A}$ define the shortened parity word

$$E(p) = 1, 0^{a_1-1} 1, 0^{a_2-1} \dots 1, 0^{a_m-1} \in \{0, 1\}^A. \quad (75)$$

If $v = (v_0, \dots, v_{A-1})$ is a binary word, set

$$N(v) = \sum_{j=0}^{A-1} v_j 2^j 3^{v_{j+1} + \dots + v_{A-1}}. \quad (76)$$

This is the numerator in Lagarias's rational-cycle formula [3, Theorem 2.1].

Theorem 4.1 (Packet-parity coordinate isomorphism). *For fixed $m \geq 1$ and $A \geq m$, the map E is a bijection*

$$E : \Pi_{m,A} \longrightarrow \mathcal{B}_{m,A}^\bullet := \{v \in \{0, 1\}^A : v_0 = 1, \sum_{j=0}^{A-1} v_j = m\}.$$

It satisfies

$$N(E(p)) = C(p).$$

Consequently the rational point attached by Lagarias's formula to $E(p)$ is exactly x_p .

Proof. The m displayed blocks in (75) have lengths $a_1, \dots, a_m$, hence total length A , and their initial symbols are exactly the m ones. Conversely, starting at the distinguished one v_0 , record the positive cyclic distances to the successive ones and finally to the boundary at A . These distances form the unique inverse packet in $\Pi_{m,A}$.

The ones of $E(p)$ occur at positions $S_0, \dots, S_{m-1}$. The number of ones strictly to the right of S_j is $m - 1 - j$. Substitution in (76) therefore gives

$$N(E(p)) = \sum_{j=0}^{m-1} 2^{S_j} 3^{m-1-j} = C(p).$$

Lagarias's denominator is $2^A - 3^m$, so the two rational points coincide. $\square$

The preceding bijection retains a distinguished odd point. Advancing that point by one odd return shifts the shortened word by a_1 positions.

Let $\sigma(a_1, \dots, a_m) = (a_2, \dots, a_m, a_1)$ and let ρ be left rotation of a binary word by one position. The packet groupoid has objects $p \in \Pi_{m,A}$ and arrows $(p, j) : p \rightarrow \sigma^j p$ indexed by $j \in \mathbb{Z}/m\mathbb{Z}$. Distinct j remain distinct arrows even when their targets coincide. The binary groupoid is the full subgroupoid of the $\mathbb{Z}/A\mathbb{Z}$ rotation action whose objects belong to $\mathcal{B}_{m,A}^\bullet$. Its arrows $(v, t) : v \rightarrow \rho^t v$ are exactly those with $v_t = 1$.

Theorem 4.2 (Exact cyclic groupoid morphism). *The object map E and arrow map*

$$(p, j) \longmapsto (E(p), S_j(p)), \quad 0 \leq j < m, \quad (77)$$

give an isomorphism from the cyclic action groupoid of packets to the pointed-one rotation groupoid of binary words. Explicitly,

$$E(\sigma^j p) = \rho^{S_j(p)} E(p). \quad (78)$$

The inverse reads the cyclic gaps between consecutive ones. In particular, p is primitive under packet rotation if and only if $E(p)$ is primitive under binary rotation, and the induced map on necklace classes is bijective.

Proof. Moving the distinguished one past the first j zero-blocks rotates the binary word by their total length $S_j(p)$, which proves (78). For composable arrows,

$$S_{j+k \bmod m}(p) \equiv S_j(p) + S_k(\sigma^j p) \pmod{A},$$

so (77) preserves composition. The positions of the ones are exactly $0 = S_0 < S_1 < \dots < S_{m-1} < A$. Thus each target arrow $(E(p), t)$ has the unique preimage (p, j) with $t = S_j$. The object inverse reads gaps, and this arrow inverse proves bijectivity on every source fibre. The identity maps to the identity. For $j > 0$,

$$S_{m-j}(\sigma^j p) = A - S_j(p),$$

so inverses are preserved as well; $j = 0$ is immediate.

Write $p = u^r$ with u primitive, of length d and sum B . Then $m = rd$, $A = rB$, and the stabilizers are

$$\text{Stab}(p) = \{kd \bmod m : 0 \leq k < r\}, \quad \text{Stab}(E(p)) = \{kB \bmod A : 0 \leq k < r\}.$$

Indeed, a stabilizing binary rotation must send the initial one to a one, hence is S_j for some j . The object bijection then forces $\sigma^j p = p$, so $d \mid j$ and $S_{kd} = kB$. Conversely all these rotations stabilize $E(p) = E(u)^r$. The arrow map sends kd to kB . In particular both stabilizers are trivial exactly when $r = 1$; this proves the primitive assertion and the bijection on cyclic classes. $\square$

This is the precise interface between the exponent necklaces used by Urata and the parity necklaces counted by Lagarias and later discussed through Lyndon words by Laarhoven–de Weger [3, 6, 5].

4.2 Cyclic transport and the exact integrality criterion

The local antecedent’s Theorem 10.1 already proves affine cyclic transport by uniqueness of the shifted fixed point. The calculation below makes the numerator identity, exact valuation, and two directions of divisibility explicit in the same coordinates.

Put $p' = (a_2, \dots, a_m)$. Directly from (74),

$$C(p) = 3^{m-1} + 2^{a_1} C(p'), \quad C(\sigma p) = 3C(p') + 2^{A-a_1}.$$

Theorem 4.3 (Cyclic affine transport). *For every positive packet p ,*

$$3C(p) + D_{m,A} = 2^{a_1} C(\sigma p), \quad 3x_p + 1 = 2^{a_1} x_{\sigma p}. \quad (79)$$

Hence

$$\nu_2(3x_p + 1) = a_1, \quad \Phi(x_p) = x_{\sigma p}.$$

For fixed m, A , divisibility $D_{m,A} \mid C(p)$ is invariant under every cyclic rotation of p .

Proof. Substitution gives

$$\begin{aligned} 3C(p) + D_{m,A} &= 3(3^{m-1} + 2^{a_1} C(p')) + 2^A - 3^m \\ &= 2^{a_1} (3C(p') + 2^{A-a_1}) = 2^{a_1} C(\sigma p). \end{aligned}$$

Division by the nonzero $D_{m,A}$ proves the rational identity. Both $C(\sigma p)$ and $D_{m,A}$ are odd, so $\nu_2(x_{\sigma p}) = 0$ and the valuation is exactly a_1 . This also proves $3x_p + 1 \neq 0$, since the numerator $C(\sigma p)$ is positive. Finally $\gcd(D_{m,A}, 6) = 1$: $D_{m,A}$ is odd and is congruent to $2^A \neq 0$ modulo 3. In the first equality, invertibility of 2^{a_1} modulo $D_{m,A}$ gives $D_{m,A} \mid C(p) \Rightarrow D_{m,A} \mid C(\sigma p)$; invertibility of 3 gives the converse. Repeating this argument handles every rotation. $\square$

Corollary 4.4 (Typed cycle criterion). *Let $p \in \Pi_{m,A}$. Then x_p is an integer if and only if $D_{m,A} \mid C(p)$; whenever it is an integer it is odd. It is positive if and only if $D_{m,A} > 0$. If, in addition, p is primitive, then $x_p \in \mathbb{Z}_{>0}$ is equivalent to a positive odd integral orbit of exact odd-step period m with exponent necklace $[p]$.*

Proof. The first assertion is the reduced rational integrality test before any cancellation: $D_{m,A} \mid C(p)$. An odd integer divided from odd $C(p)$ by odd $D_{m,A}$ is odd. Since $C(p) > 0$, its sign is the sign of $D_{m,A}$. Theorem 4.3 supplies every actual odd-step arrow. A shorter orbit period would make the exponent itinerary repeat, contradicting primitivity. Conversely, for $f_a(x) = (3x+1)/2^a$, induction on the number of branches gives

$$(f_{a_m} \circ \dots \circ f_{a_1})(x) = \frac{3^m x + C(p)}{2^A}.$$

At an actual return this forces $D_{m,A}x = C(p)$, hence $x = x_p$. If the itinerary were a proper power, its primitive root would itself supply this periodic point by the same affine identity, contradicting the assumed exact period. $\square$

Theorem 4.5 (Exact comparison of three clocks). *Let $p \in \Pi_{m,A}$ and $x_j = x_{\sigma^j p}$. For $0 \leq j < m$,*

$$\Phi^j(x_p) = T^{S_j}(x_p) = F^{S_j+j}(x_p) = x_j.$$

Within its $(j+1)$ st block the intermediate values are exactly

$$\begin{aligned} T^s(x_j) &= 2^{a_{j+1}-s}x_{j+1} & (1 \leq s \leq a_{j+1}), \\ F^s(x_j) &= 2^{a_{j+1}+1-s}x_{j+1} & (1 \leq s \leq a_{j+1} + 1). \end{aligned}$$

The full parity word is

$$E_F(p) = 1, 0^{a_1} \dots 1, 0^{a_m} \in \{0, 1\}^{A+m}.$$

If $p = u^r$ with u primitive of length d and sum B , the exact periods of x_p under Φ, T, F are respectively $d, B, B + d$.

Proof. The transport identity is $3x_j + 1 = 2^{a_{j+1}}x_{j+1}$, with x_{j+1} odd and nonzero. The first shortened step gives $2^{a_{j+1}-1}x_{j+1}$; successive halving steps give the displayed T formula. Its values are even before the last and odd at the last. The full map inserts $2^{a_{j+1}}x_{j+1}$ as its first intermediate value, proving the F formula and its parity block. Concatenation gives all return times. Thus E_F inserts one zero after each one of E ; its inverse deletes that zero. These are inverse bijections between $\mathcal{B}_{m,A}^\bullet$ and the pointed binary words of length $A + m$, weight m , having at least one zero between consecutive cyclic ones.

To establish exact periods, put $f_u = f_{u_d} \circ \dots \circ f_{u_1}$. It has positive slope $\alpha = 3^d/2^B \neq 1$, and its r th iterate is f_p . Since

$$f_u^r(x) - x = (1 + \alpha + \dots + \alpha^{r-1})(f_u(x) - x),$$

the two maps have the same unique fixed point: $x_p = x_u$. The least odd-return period is d , since a shorter period would repeat the exponent itinerary of u . Any return to the odd starting point under T or F must be one of the odd returns just enumerated. Their first return to x_u occurs after the d blocks of u , at times B and $B + d$. This proves the three exact periods, including imprimitive packets. $\square$

4.3 Fixed-content primitive necklaces and the exact sign split

For $m \geq 1$ and $A \geq m$, let $W(m, A)$ be the number of words in $\Pi_{m,A}$ whose least cyclic period is m . Let $N_{\text{orb}}^+(m, A)$ and $N_{\text{orb}}^-(m, A)$ denote the numbers of positive and negative rational Φ -orbits of exact odd-return period m represented by this packet, modulo cyclic phase. Finally, let $N_{\text{pt}}^\pm(m, A)$ count the individual rational points in those orbits. These are three different counts: primitive pointed words, cyclic orbits, and orbit points.

Theorem 4.6 (Signed fixed-content necklace count). *Put $g = \gcd(m, A)$. Then*

$$W(m, A) = \sum_{d|g} \mu(d) \binom{A/d-1}{m/d-1}. \quad (80)$$

The unsigned orbit count has the two exactly equal forms

$$N_{\text{orb}}(m, A) = \frac{1}{m} \sum_{d|g} \mu(d) \binom{A/d-1}{m/d-1} = \frac{1}{A} \sum_{d|g} \mu(d) \binom{A/d}{m/d}. \quad (81)$$

Its sign decomposition is

$$N_{\text{orb}}^+(m, A) = \mathbf{1}_{\{2^A > 3^m\}} N_{\text{orb}}(m, A), \quad N_{\text{orb}}^-(m, A) = \mathbf{1}_{\{2^A < 3^m\}} N_{\text{orb}}(m, A), \quad (82)$$

$$N_{\text{pt}}^+(m, A) = m N_{\text{orb}}^+(m, A), \quad N_{\text{pt}}^-(m, A) = m N_{\text{orb}}^-(m, A). \quad (83)$$

There is no zero-sign case. If $\gcd(m, A) = 1$, every word in $\Pi_{m,A}$ is primitive and therefore

$$N_{\text{orb}}(m, A) = \frac{1}{m} \binom{A-1}{m-1}, \quad N_{\text{pt}}(m, A) = \binom{A-1}{m-1}. \quad (84)$$

Proof. There are $\binom{A-1}{m-1}$ positive compositions of A into m parts. Every such word has a unique primitive root. If it is the d th power of a primitive word, then $d \mid m$ and $d \mid A$, and the root has length m/d and exponent sum A/d . Conversely, repeating any primitive word with those parameters d times recovers a word in $\Pi_{m,A}$. Consequently

$$\binom{A-1}{m-1} = \sum_{d \mid g} W(m/d, A/d).$$

Möbius inversion on the common divisors of (m, A) gives (80); no Burnside coefficient remains hidden in this inversion.

The cyclic action on a primitive word is free: a nonidentity stabilizing rotation would exhibit the word as a proper power. Each primitive necklace therefore contains exactly m pointed words. Theorems 4.2 and 4.5 identify one such necklace with one rational Φ -orbit of exact period m . Conversely, the orbit determines its successive exact valuations, so two necklaces cannot determine the same orbit. This proves the first form of (81). For each $d \mid g$,

$$\frac{1}{m} \binom{A/d-1}{m/d-1} = \frac{1}{A} \binom{A/d}{m/d},$$

which proves the fixed-Hamming-weight binary-necklace form.

For every rotation $\sigma^j p$, equation (74) gives

$$x_{\sigma^j p} = \frac{C(\sigma^j p)}{2^A - 3^m}, \quad C(\sigma^j p) > 0.$$

Thus all m points of the orbit have the sign of $2^A - 3^m$. Equality $2^A = 3^m$ is impossible by unique factorization. This proves (82); exact period supplies the m distinct points in (83). Finally, a proper power would give a repetition number dividing both m and A , so coprimality forces every word to be primitive and yields (84). $\square$

Stanley's multivariate Lyndon series counts primitive necklaces by the number of occurrences of each colour [4, Exercise 7.89(a), equations (7.149)–(7.150), and the solution on printed p. 555]. Extracting the binary-content coefficient with content $(m, A - m)$ gives exactly the second expression in (81). Lagarias gives the unrestricted primitive binary-necklace and rational-cycle coordinates, while Laarhoven–de Weger restate the unrestricted count in Lyndon-word language [3, pp. 38–39][6, lines 303–315]. Thus the enumeration formula is classical. The packet-parity groupoid proves its exact transport to exponent compositions, and the denominator $2^A - 3^m$ supplies the Collatz-specific sign partition. Summing over $1 \leq m \leq A$ recovers the unrestricted primitive binary-necklace count for every $A \geq 2$. The sign refinement then uses the packet

denominator $2^A - 3^m$ and does not alter the classical necklace enumeration. At the boundary, $m = 1$ gives the negative fixed point for $A = 1$ and one positive fixed point for every $A \geq 2$, whereas $A = m > 1$ contains only the imprimitive word $(1, \dots, 1)$ and contributes no exact-period- m orbit.

4.4 The exact minimum strip and its literature boundary

Let $(x_0, \dots, x_{m-1})$ be a positive odd integral cycle and write $2^{a_i} x_{i+1} = 3x_i + 1$, with indices modulo m . Multiplication around the cycle gives the exact identity

$$2^A = \prod_{i=0}^{m-1} \left(3 + \frac{1}{x_i} \right). \quad (85)$$

This product identity and its minimum-point use are classical; compare Crandall's cycle formula and minimum estimate [2, Section 7].

If the cycle is nonconstant and $n = \min_i x_i$, then

$$3^m < 2^A < \left(3 + \frac{1}{n} \right)^m. \quad (86)$$

The left inequality is strict termwise. The right inequality is strict because at least one x_i exceeds n . A constant positive integral cycle would satisfy $x = 1/(2^a - 3)$ and is therefore the fixed point $x = 1$, $a = 2$. Moreover 1 is fixed, while $3 \mapsto 5 \mapsto 1$ under Φ ; hence a nontrivial positive cycle has $n \geq 7$. Thus

$$3^m < 2^A < \left(\frac{22}{7} \right)^m. \quad (87)$$

No logarithmic approximation is required to enumerate the integral points in this strip: test

$$3^m < 2^A \quad \text{and} \quad 7^m 2^A < 22^m$$

as integer inequalities. For $2 \leq m \leq 17$ the result is

m	admissible A	m	admissible A
2, 3, 4	none	10	16
5	8	11	18
6, 7	none	12	none
8	13	13	21
9	none	14	23
		15	24
		16	26
		17	27, 28

Remark 4.7 (Historical comparison). Crandall obtained a much larger cycle-length exclusion in 1978 from the verification range available at that time [2, Theorem 7.3]. The small table here supplies explicit packet labels for the residue calculations below.

4.5 Finite-state residue recursion

For a positive odd modulus q , define

$$R_{m,A}(q) = \{C(p) \bmod q : p \in \Pi_{m,A}\} \subseteq \mathbb{Z}/q\mathbb{Z}.$$

Writing $p = (a, p')$ gives

$$C(a, p') = 3^{m-1} + 2^a C(p'). \quad (88)$$

Theorem 4.8 (Packet residue automaton). *For $m \geq 2$ and $A \geq m$,*

$$R_{m,A}(q) = \bigcup_{a=1}^{A-m+1} \left(3^{m-1} + 2^a R_{m-1,A-a}(q) \right), \quad R_{1,A}(q) = \{\bar{1}\} \quad (A \geq 1). \quad (89)$$

Let

$$v_{m,A}^{(q)}(r) = \#\{p \in \Pi_{m,A} : C(p) \equiv r \pmod{q}\}.$$

For $A < m$ set $R_{m,A}(q) = \emptyset$ and $v_{m,A}^{(q)} = 0$. With column vectors, define

$$M_{m,a}^{(q)} e_r = e_{3^{m-1}+2^a r}, \quad (M_{m,a}^{(q)})^{-1} e_s = e_{2^{-a}(s-3^{m-1})},$$

where all indices lie in $\mathbb{Z}/q\mathbb{Z}$. Put $\lambda = \text{ord}_q(2)$ for $q > 1$, and $\lambda = 1$ for $q = 1$. Then the formal vector series $V_m^{(q)}(z) = \sum_{A \geq m} v_{m,A}^{(q)} z^A$ obeys

$$V_1^{(q)}(z) = \frac{z}{1-z} e_{\bar{1}}, \quad V_m^{(q)}(z) = \frac{\sum_{a=1}^{\lambda} z^a M_{m,a}^{(q)}}{1-z^\lambda} V_{m-1}^{(q)}(z). \quad (90)$$

Every coordinate of $V_m^{(q)}$ is therefore rational in z .

Proof. Every packet has a unique first part $1 \leq a \leq A - m + 1$ and a unique tail in $\Pi_{m-1,A-a}$. Equation (88) proves the set recursion and, with multiplicities retained, the vector recursion

$$v_{m,A}^{(q)} = \sum_{a=1}^{A-m+1} M_{m,a}^{(q)} v_{m-1,A-a}^{(q)}.$$

Since q is odd, multiplication by 2^a is invertible modulo q , so each $M_{m,a}^{(q)}$ is a permutation matrix. The equality $2^{a+\lambda} \equiv 2^a \pmod{q}$ makes the matrix sequence λ -periodic. Each $a \geq 1$ has a unique form $b + k\lambda$, $1 \leq b \leq \lambda$, $k \geq 0$. In $\mathbb{Z}^q[[z]]$, every coefficient of the following product is a finite sum, and this decomposition gives

$$\sum_{a \geq 1} z^a M_{m,a}^{(q)} = \frac{\sum_{b=1}^{\lambda} z^b M_{m,b}^{(q)}}{1-z^\lambda}.$$

This proves (90). More explicitly, putting $P_j(z) = \sum_{b=1}^{\lambda} z^b M_{j,b}^{(q)}$ gives

$$V_m^{(q)}(z) = \frac{P_m(z) P_{m-1}(z) \cdots P_2(z) z e_{\bar{1}}}{(1-z)(1-z^\lambda)^{m-1}}.$$

The order of the matrix factors is as displayed. For $m = 1$ their product is the identity. When $q = 1$ the formula reduces to $V_m(z) = z^m/(1-z)^m$, the generating function for positive compositions. $\square$

Corollary 4.9 (Divisor exclusion). *If q is a positive divisor of $|D_{m,A}|$ and $0 \notin R_{m,A}(q)$, then $D_{m,A} \nmid C(p)$ for every $p \in \Pi_{m,A}$; hence the packet contains no integral periodic point.*

Proof. The divisibility $D_{m,A} \mid C(p)$ would imply $q \mid C(p)$, contradicting $0 \notin R_{m,A}(q)$. $\square$

4.6 Synchronized residues and the Chinese remainder map

Factorwise information must retain the exponent word that produced each residue. The common-letter product of weighted automata is the standard construction behind the Hadamard product of recognizable series [22, Definition 4.1.4, Proposition 4.3.10]. Here its coordinate map can be written explicitly.

Theorem 4.10 (Synchronized CRT transport). *Let q_1, q_2 be coprime positive odd integers and $q = q_1 q_2$. For each modulus h , let L_h be the free abelian group with basis $\{e_r : r \in \mathbb{Z}/h\mathbb{Z}\}$. The map*

$$J : L_q \longrightarrow L_{q_1} \otimes_{\mathbb{Z}} L_{q_2}, \quad J e_r = e_{r \bmod q_1} \otimes e_{r \bmod q_2}$$

is an isomorphism preserving nonnegative coefficients. Choose integers c_1, c_2 with $q_2 c_1 \equiv 1 \pmod{q_1}$ and $q_1 c_2 \equiv 1 \pmod{q_2}$. Its inverse on basis vectors is

$$J^{-1}(e_u \otimes e_v) = e_{q_2 c_1 u + q_1 c_2 v \bmod q}.$$

For every $m \geq 2$ and $a \geq 1$,

$$J M_{m,a}^{(q)} = (M_{m,a}^{(q_1)} \otimes M_{m,a}^{(q_2)}) J. \quad (91)$$

In particular, if

$$w_{m,A}(u, v) = \#\{p \in \Pi_{m,A} : C(p) \equiv u \pmod{q_1}, C(p) \equiv v \pmod{q_2}\},$$

then $w_{m,A} = J v_{m,A}^{(q)}$ and

$$w_{m,A} = \sum_{a=1}^{A-m+1} (M_{m,a}^{(q_1)} \otimes M_{m,a}^{(q_2)}) w_{m-1, A-a}. \quad (92)$$

The base vector is $w_{1,A} = e_{\bar{1}} \otimes e_{\bar{1}}$ for $A \geq 1$, and $w_{m,A} = 0$ when $A < m$.

Proof. Reducing $q_2 c_1 u + q_1 c_2 v$ modulo either factor gives u or v , respectively. If two integers have the same pair of residues, their difference is divisible by both coprime factors and hence by q . These facts prove both inverse identities, including independence from the chosen representatives and Bezout coefficients. Thus J permutes the basis, has zero kernel and singleton fibres, and preserves each coefficient.

Both sides of (91) send e_r to

$$e_{3^{m-1} + 2^a r \bmod q_1} \otimes e_{3^{m-1} + 2^a r \bmod q_2}.$$

This proves the identity and, by composition, the corresponding identity for every sequence of first-letter transitions. Summing the basis image over the same packets proves $w_{m,A} = J v_{m,A}^{(q)}$. Applying J to the scalar-modulus vector recursion proves (92); no independent choice of letters in the two factors occurs. The base and empty ranges follow from their definitions. $\square$

The factorwise inverse transition remains $r \mapsto 2^{-a}(r - 3^{m-1})$. Consequently this comparison loses no residue information. If λ is the least common multiple of the two orders of 2 (with the singleton convention for modulus one), the joint generating series $W_m(z) = \sum_{A \geq m} w_{m,A} z^A$ satisfies

$$W_m(z) = \frac{\sum_{a=1}^{\lambda} z^a (M_{m,a}^{(q_1)} \otimes M_{m,a}^{(q_2)})}{1 - z^{\lambda}} W_{m-1}(z), \quad W_1(z) = \frac{z}{1 - z} e_{\bar{1}} \otimes e_{\bar{1}}.$$

The proof is the same unique decomposition $a = b + k\lambda$ used above; every coefficient is a finite sum. Notice the single power z^a and the same letter a in both factors. A product of the separately summed generating series would instead select two words and two budgets. For any fixed m, A , the alphabet $1, \dots, A$ and the length and budget coordinates give finite layered automata. Start at $(j, B, r) = (0, 0, 0)$, read the reversed word $a_m, \dots, a_1$, and use

$$(j, B, r) \mapsto (j + 1, B + a, 3^j + 2^a r \bmod q), \quad j < m, \quad B + a \leq A.$$

Accept at $j = m, B = A$. The first transition sets the residue to 1; subsequent transitions prepend a letter. The synchronized product shares j, B, a and retains the pair of residues. It is the common-letter construction cited above, restricted to the shared length and budget layers.

Proposition 4.11 (Information discarded by the marginals). *On integer residue tables define*

$$P(w) = \left(\left(\sum_v w_{uv} \right)_u, \left(\sum_u w_{uv} \right)_v \right).$$

Its image consists of pairs of integer vectors with equal total sum. Its kernel has the basis

$$E_{uv} - E_{u0} - E_{0v} + E_{00} \quad (u \neq 0, v \neq 0),$$

where $E_{uv} = e_u \otimes e_v$, and has rank $(q_1 - 1)(q_2 - 1)$. Each nonempty fibre is a translate of this kernel; for nonnegative count tables it is the intersection of that translate with the nonnegative cone.

Proof. The two marginal totals both equal $\sum_{u,v} w_{uv}$. Given marginals α, β with common total N , the integer table $\sum_u \alpha_u E_{u0} + \sum_v \beta_v E_{0v} - N E_{00}$ has those marginals. Every displayed rectangle has zero row and column sums. If w has zero marginals, subtract the sum of those rectangles with coefficients w_{uv} for $u, v \neq 0$. The remaining table is supported on row zero and column zero, and its zero marginals force every remaining entry to vanish. Independence follows by inspecting each (u, v) entry with $u, v \neq 0$. The fibre statements follow by subtracting two preimages. $\square$

Proposition 4.12 (Reduced denominators and integral $3x + d$ cycles). *Let $p \in \Pi_{m,A}$ with $D = 2^A - 3^m > 0$, and set $g = \gcd(C(p), D)$ and $d = D/g$. Then g is invariant under cyclic rotation, and*

$$n_j = \frac{C(\sigma^j p)}{g} = d x_{\sigma^j p}$$

are positive odd integers satisfying $\gcd(n_j, d) = 1$ and

$$3n_j + d = 2^{a_{j+1}} n_{j+1}, \quad \nu_2(3n_j + d) = a_{j+1}.$$

Multiplication by d conjugates the rational shortened orbit to the integral orbit of

$$T_d(n) = \begin{cases} (3n + d)/2, & n \text{ odd}, \\ n/2, & n \text{ even}. \end{cases}$$

Its inverse on these orbits is division by d . For primitive p , the odd-return and shortened periods are exactly m and A .

Proof. The identity $3C(p) + D = 2^{a_1}C(\sigma p)$ and $\gcd(D, 6) = 1$ give $\gcd(C(p), D) = \gcd(C(\sigma p), D)$ by taking gcds with D . Division by this common odd g proves the displayed recurrence; its right side has odd nonzero n_{j+1} , proving the valuation. Reduction by the full gcd gives $\gcd(n_j, d) = 1$. Since d is odd, scaling by d preserves parity, and checking the two branches gives $T_d(dx) = dT(x)$ at every point of the rational orbit, including its even intermediate points. Division by d proves both inverse identities and preserves all return times. The exact periods follow from Theorem 4.5. $\square$

The source comparison here is Belaga–Mignotte’s configuration-to-membership map [23, Sections 7–8]: their $(k, \ell, a(P), B_{k,\ell}, H, F, G)$ correspond respectively to $(m, A, C(p), D, g, n_0, d)$. Their aperiodic configuration is a primitive exponent word; their primitive $3x + d$ cycle instead specifies coprimality with d . The two conditions serve different purposes in this correspondence.

4.7 A complete joint-residue calculation at $(m, A) = (10, 16)$

Here

$$D_{10,16} = 2^{16} - 3^{10} = 6487 = 13 \cdot 499, \quad |\Pi_{10,16}| = \binom{15}{9} = 5005.$$

The CRT inverse is particularly explicit:

$$(u, v) \mapsto 3992u + 2496v \pmod{6487}, \quad 3992 = 499 \cdot 8, \quad 2496 = 13 \cdot 192.$$

The first coefficient is $(1, 0)$ and the second $(0, 1)$ in the two residue coordinates. Direct word enumeration and the independent joint recursion give the following exact stratification:

$\gcd(C(p), 6487)$	all words	primitive words	primitive cyclic classes
1	4560	4560	456
13	410	410	41
499	35	0	0
6487	0	0	0
Total	5005	4970	497

There are 3473 occupied joint cells and 3014 zero cells. Both marginal supports are full, but the marginal zero counts 410 and 35 have joint zero count $w_{10,16}(0, 0) = 0$. Thus a word realizing zero in one factor and a word realizing zero in the other need not be the same word.

For explicit witnesses,

p	$C(p)$	$C(p) \bmod 13$	$C(p) \bmod 499$
$(1, 1, 1, 1, 1, 1, 2, 2, 2, 4)$	65065	0	195
$(1, 1, 1, 1, 4, 1, 1, 1, 1, 4)$	105289	2	0

The first word is primitive; the second is a square. In fact every proper repetition number divides both 10 and 16, so every imprimitive word is $p = u^2$ for $u \in \Pi_{5,8}$. There are exactly 35 such words. Expanding the numerator in its first and second five-term blocks gives

$$C(u^2) = (3^5 + 2^8)C(u) = 499C(u), \quad D_{10,16} = (2^8 - 3^5)(2^8 + 3^5) = 13 \cdot 499.$$

The table shows that these are all the 499-divisible words. Therefore 499 alone rules out an integral periodic point for every primitive packet at this label, although it does not exclude zero

residues for all positive words. The 35 squares form seven classes of least odd period five, in addition to the 497 primitive classes of period ten. The count of all cyclic classes is consequently 504.

Under Proposition 4.12, the first two rows give 456 coprime $3x + 6487$ cycles and 41 coprime $3x + 499$ cycles, each with ten odd returns and sixteen shortened steps. The third row gives seven coprime $3x + 13$ cycles with five odd returns and eight shortened steps, represented here twice. The 456 count is already reported by Belaga–Mignotte [23, Example 8.5(1)]; the 4560 pointed words are its ten choices of odd base point per cycle.

These assertions are finite exact results: the supplied certificate checks every one of the 5005 words and every cell of the 13-by-499 table. Its joint recurrence uses the same first letter in both coordinates, and its row and column sums agree with separately computed scalar recurrences.

4.8 Exact finite packet certificates through period sixteen

For $(m, A) = (5, 8)$, $D_{5,8} = 13$. Since $\gcd(5, 8) = 1$, all 35 packets are primitive and form seven cyclic classes. The lexicographically minimal representatives give:

p	$C(p)$	$C(p) \bmod 13$
(1, 1, 1, 1, 4)	211	3
(1, 1, 1, 2, 3)	227	6
(1, 1, 1, 3, 2)	259	12
(1, 1, 2, 1, 3)	251	4
(1, 1, 2, 2, 2)	283	10
(1, 1, 3, 1, 2)	331	6
(1, 2, 1, 2, 2)	319	7

Theorem 4.3 makes divisibility invariant on each class, so the packet has no integral point.

For $(m, A) = (8, 13)$,

$$D_{8,13} = 2^{13} - 3^8 = 1631 = 7 \cdot 233.$$

Here $|\Pi_{8,13}| = \binom{12}{7} = 792$ and coprimality of 8 and 13 makes every packet primitive, hence there are $792/8 = 99$ cyclic classes. The recursion (89) gives the exact set

$$R_{8,13}(233) = \mathbb{Z}/233\mathbb{Z} \setminus \{0, 138\}. \quad (93)$$

Among the 99 minimal representatives, precisely 15 have numerator divisible by 7, none has numerator divisible by 233, and none is integral. Thus the factor 233 excludes the entire packet.

The exact strip leaves the following candidate packets through period sixteen. Two independent streaming enumerations—separator combinations with the displayed sum for C , and depth-first compositions with $B_{j+1} = 3B_j + 2^{S_j}$ —produce identical word streams and the following counts.

m	A	$ \Pi_{m,A} $	$D_{m,A}$	integral packets
5	8	35	13	0
8	13	792	1631	0
10	16	5005	6487	0
11	18	19448	84997	0
13	21	125970	502829	0
14	23	497420	3605639	0
15	24	817190	2428309	0
16	26	3268760	24062143	0

Each method processes exactly 4,734,620 words. In the packets (5, 8), (8, 13), (11, 18), and (13, 21), the respective prime factors

$$13, \quad 233, \quad 7727, \quad 502829$$

already exclude zero numerator residues.

Theorem 4.13 (Certified finite consequence). *There is no nontrivial positive odd integral periodic orbit of exact odd-step period $2 \leq m \leq 16$.*

Proof. For $m = 2, 3, 4, 6, 7, 9, 12$, the exact strip (87) contains no integer A . The remaining values 5, 8, 10, 11, 13, 14, 15, 16 have exactly the packets displayed in the table, and each exhaustive divisibility count is zero. The packet criterion in Corollary 4.4 completes the proof. $\square$

The same strip first yields two packet labels at $m = 17$, namely (17, 27) and (17, 28). This is only the next finite locus selected by (87); it is not asserted to be the frontier of the published cycle literature.

4.9 Reproducible calculations

The portable script `certificates/affine_packet_residue_checks.py` uses no floating point. It checks the packet-parity numerator identity and all three clocks on 12,089 packets: 155,047 shortened steps, 233,525 full steps, and 78,478 odd returns. These include 11,989 primitive and 100 imprimitive packets; the first return is checked in each clock. On a separate 461-packet range it checks the groupoid arrows, inverses, composition, and stabilizers in both binary clocks. Direct enumeration checks 702 residue multiplicity vectors, including 189 empty ranges $A < m$; 546 coefficient vectors independently check the rational-series recursion for the odd moduli 1, 3, 5, 7, 9, 11, 13. It computes (93) both by direct enumeration and by (89). Finally it runs the two independent full scans above and pins every stream hash, minimum numerator, maximum numerator, denominator, and count. Optimized Python is rejected so that assertions cannot be disabled.

The exhaustive scans stop at period sixteen; the two period-seventeen labels are not included. The general coordinate and generating-series results are proved above independently of these finite checks.

The additional script `certificates/affine_crt_synchronization_checks.py` independently checks the joint-residue calculation at (10, 16). Two word generators and three numerator formulas agree; it checks all 6487 CRT inverse pairs, 408681 transition/inverse instances, the 35 repeated-root identities, and the full joint table including zero cells. The certificate rejects optimized Python and can write a new execution receipt without overwriting an existing one.

5 Packet fixed-point series and their dyadic poles

The affine coordinates retain useful analytic information when summed over all words of a prescribed length and exponent sum. The total affine offset has a rational generating function, whereas the total fixed point has infinitely many poles. We construct both functions in the original packet coordinates and determine their exact domains of power-series convergence. The rational recursion and formal dilation equation occur in the local *Affine Packets* note, Sections 15–17, and in raw Chatnotes *Implications of Affine Formula*, lines 432–475. The continuation below extends the note’s one-letter pole calculation to every packet length. These statements concern

sums over rational fixed points, with repeated words retained; they are not sums over positive integral cycles alone.

5.1 The rational offset family

For $m \geq 1$ and $A \geq m$, retain $C(p)$, $D = 2^A - 3^m$, and $x_p = C(p)/D$ from (74). Put

$$c(p) = 2^{-A}C(p), \quad G_m(A) = |\Pi_{m,A}|, \quad S_m(A) = \sum_{p \in \Pi_{m,A}} c(p), \quad X_m(A) = \sum_{p \in \Pi_{m,A}} x_p.$$

Set these three coefficients to zero for $A < m$. Their generating functions are $G_m(q) = \sum_A G_m(A)q^A$, $F_m(q) = \sum_A S_m(A)q^A$, and $X_m(q) = \sum_A X_m(A)q^A$. The same letter X_m with an integer argument denotes a coefficient; with q it denotes the series. Define

$$u(q) = \frac{q}{1-q}, \quad v(q) = \frac{q}{2-q}.$$

Proposition 5.1 (Packet sums and their two-state representation). *As formal series, and hence as rational functions where defined,*

$$G_m(q) = u(q)^m, \tag{94}$$

$$F_{m+1}(q) = v(q)(3F_m(q) + G_m(q)), \quad F_1(q) = v(q), \tag{95}$$

$$F_m(q) = \sum_{k=0}^{m-1} 3^{m-1-k} u(q)^k v(q)^{m-k}. \tag{96}$$

In particular, with column vectors,

$$\begin{pmatrix} F_{m+1} \\ G_{m+1} \end{pmatrix} = \begin{pmatrix} 3v & v \\ 0 & u \end{pmatrix} \begin{pmatrix} F_m \\ G_m \end{pmatrix}, \quad \begin{pmatrix} F_1 \\ G_1 \end{pmatrix} = \begin{pmatrix} v \\ u \end{pmatrix}.$$

The bivariate formal series is

$$\sum_{m \geq 1} F_m(q) z^{m-1} = \frac{v(q)}{(1 - 3v(q)z)(1 - u(q)z)}. \tag{97}$$

For $m \geq 2$, F_m has exactly two poles: order $m - 1$ at 1 and order m at 2. For $m = 1$ its only pole is the simple pole at 2. At the regular point $q = 1/2$, $F_m(1/2) = m/3$.

Proof. A word is a composition of A into m positive parts, so $G_m(A) = \binom{A-1}{m-1}$ and its series is u^m . Appending $a \geq 1$ to a word p gives $c(p, a) = 2^{-a}(3c(p) + 1)$. Sum first over $p \in \Pi_{m, A-a}$ and then over $1 \leq a \leq A - m$. Each coefficient is a finite sum, and multiplication by $\sum_{a \geq 1} 2^{-a} q^a = v(q)$ gives (95). Induction proves (96). Summing that finite-sum formula over m gives the product of the two geometric series in (97); both denominators have constant term one in z .

At $q = 1$, only the $k = m - 1$ term of (96) has pole order $m - 1$, and its coefficient relative to $(1 - q)^{-(m-1)}$ is one. At $q = 2$, only the $k = 0$ term has order m , and its coefficient relative to $(2 - q)^{-m}$ is $3^{m-1}2^m$. Both coefficients are nonzero. The finite-sum formula has no other denominator zeros. Finally $u(1/2) = 1$ and $v(1/2) = 1/3$, so every one of the m terms in the finite sum equals $1/3$. $\square$

5.2 Formal inversion and convergence domains

Proposition 5.2 (The exact dilation map). *For each $m \geq 1$, the map*

$$\mathcal{D}_m : q^m \mathbb{C}[[q]] \longrightarrow q^m \mathbb{C}[[q]], \quad f(q) \longmapsto f(2q) - 3^m f(q)$$

is a continuous complex-linear bijection for the q -adic topology. Its inverse is

$$\mathcal{D}_m^{-1} \left(\sum_{A \geq m} h_A q^A \right) = \sum_{A \geq m} \frac{h_A}{2^A - 3^m} q^A.$$

Both maps preserve the order of every nonzero formal series. Their kernels are zero and every fibre is a singleton. In particular,

$$X_m(2q) - 3^m X_m(q) = F_m(2q). \quad (98)$$

The radius of convergence of X_1 is 2. For every $m \geq 2$ the radius of X_m is exactly 1. Consequently (98), interpreted as an equality of the defining convergent power series, holds on $|q| < 1$ for $m = 1$, and on $|q| < 1/2$ for $m \geq 2$.

Proof. The coefficient multiplier $2^A - 3^m$ is nonzero: its first term is even and its second odd. The two displayed maps are inverse coefficient by coefficient and preserve the least index of a nonzero coefficient. This proves all the formal topological and linear assertions. It does not assert preservation of multiplication. The identity

$$X_m(A) = \frac{S_m(A)}{1 - 3^m 2^{-A}}$$

then gives (98).

For $m = 1$, $X_1(A) = (2^A - 3)^{-1}$ and $|X_1(A)|^{1/A} \rightarrow 1/2$. For fixed m , every word satisfies

$$0 < c(p) \leq \sum_{j=0}^{m-1} 3^{m-1-j} 2^{-(m-j)} = (3/2)^m - 1.$$

Here $A - S_j \geq m - j$ because the remaining letters are positive. For all sufficiently large A , $1 - 3^m 2^{-A} \geq 1/2$. Therefore

$$0 < X_m(A) \leq 2((3/2)^m - 1) \binom{A-1}{m-1}.$$

The finitely many earlier coefficients do not affect the radius. This polynomial upper bound gives convergence for $|q| < 1$. For $m \geq 2$ and each $A \geq m$, there is a word whose last letter is 1, for example $(A - m + 1, 1, \dots, 1)$. Its final numerator summand gives $c(p) \geq 1/2$. Once $2^A > 3^m$, all packet terms are positive, so $X_m(A) \geq 1/2$. The terms therefore cannot tend to zero when $|q| > 1$. The radius is exactly one, and the claimed domains follow after replacing q by $2q$. The same lower bound shows failure of the term test at every point of the boundary circle for $m \geq 2$. For $m = 1$, the terms on $|q| = 2$ have magnitudes $2^A/|2^A - 3| \rightarrow 1$, so they also fail that test. $\square$

For example,

$$X_2(A) = \frac{2^A + 3A - 5}{2^A - 9} \quad (A \geq 2),$$

because $C(a, A - a) = 3 + 2^a$. Thus $X_2(A) \rightarrow 1$. At $q = 3/4$, the terms defining $X_2(2q)$ do not tend to zero. This disproves the whole-unit-disc convergence assertion appended to the formal identity in the local note's Theorem 17.1. The formal identity itself is exact.

5.3 Meromorphic continuation with exact pole orders

Theorem 5.3 (The dyadic pole family). *For every $m \geq 1$, X_m has a meromorphic continuation to the complex plane. When $m = 1$ its poles are precisely 2^j , $j \geq 1$, all simple. When $m \geq 2$, it has a pole of order $m - 1$ at 1 and a pole of order m at each 2^j , $j \geq 1$, with no other poles. More precisely,*

$$\lim_{q \rightarrow 1} (1 - q)^{m-1} X_m(q) = 1 \quad (m \geq 2), \quad (99)$$

$$\lim_{q \rightarrow 2^j} (2^j - q)^m X_m(q) = 3^{mj-1} 2^{jm} \quad (m \geq 1, j \geq 1). \quad (100)$$

In particular, none of the functions X_m is rational.

Proof. Choose an integer $L \geq m$ with $2^L > 3^m$; for example $L = 2m$. Set

$$P_m(q) = \sum_{A=m}^{L-1} X_m(A) q^A, \quad H_m(q) = F_m(q) - \sum_{A=m}^{L-1} S_m(A) q^A.$$

The empty sums are zero. The rational function H_m vanishes to order at least L at zero and has exactly the same poles and principal parts as F_m . The following explicit expression continues X_m :

$$X_m(q) = P_m(q) + \sum_{r \geq 0} 3^{mr} H_m(q/2^r). \quad (101)$$

Indeed, for $A \geq L$,

$$\frac{1}{1 - 3^m 2^{-A}} = \sum_{r \geq 0} 3^{mr} 2^{-rA}.$$

For $|q| < 1$, the sum of absolute values of this double series is at most

$$\frac{1}{1 - 3^m 2^{-L}} \sum_{A \geq L} S_m(A) |q|^A < \infty.$$

The previous proposition supplies the polynomial bound needed here. Interchanging the sums proves (101) near zero.

For an explicit normal-convergence estimate put

$$B_{m,L} = 2^L H_m(1/2) > 0, \quad \eta = 3^m 2^{-L} < 1.$$

All coefficients of H_m from degree L onward are positive. Hence $|H_m(z)| \leq B_{m,L} |z|^L$ for $|z| \leq 1/2$. For $|q| \leq R$ and any integer $N \geq 0$ with $R/2^N \leq 1/2$,

$$\sup_{|q| \leq R} \left| \sum_{r=N}^{\infty} 3^{mr} H_m(q/2^r) \right| \leq \frac{B_{m,L} R^L \eta^N}{1 - \eta}. \quad (102)$$

The constant is explicitly rational: $B_{m,L} = 2^L (m/3 - \sum_{A=m}^{L-1} S_m(A) 2^{-A})$. This is a summable geometric bound uniform on every compact set. Only finitely many earlier summands remain. Each is rational, with poles confined to the positive dyadic points. Away from those points, the sum is locally uniformly convergent and holomorphic. Near each such point, all but finitely many summands are holomorphic and their tail converges uniformly, so the sum is meromorphic there.

At $q = 1$ only $r = 0$ can be singular, giving (99). At $q = 2^j$, $j \geq 1$, the term $r = j - 1$ inherits the pole of F_m at 2, of order m . Its leading coefficient in powers of $(2^j - q)^{-1}$ is

$$3^{m(j-1)} 3^{m-1} (2^j)^m = 3^{mj-1} 2^{jm}.$$

For $m \geq 2$, the only other singular term is $r = j$, whose order is $m - 1$; it cannot cancel the leading pole. For $m = 1$ that term is regular. No other terms are singular. This proves the exact pole list and both limits. A rational function has only finitely many poles, whereas this list is infinite. $\square$

The same expression also proves that the dilation equation holds globally as an identity of meromorphic functions, by analytic continuation from its disc of convergent series. This is a different assertion from convergence of the defining series beyond its radius. For $m = 1$, taking $L = 2$ gives $P_1(q) = -q$ and recovers the source's formula

$$X_1(q) = -q + \sum_{r \geq 0} 3^r \frac{(q/2^{r+1})^2}{1 - q/2^{r+1}}.$$

Thus the one-letter pole calculation and the general packet continuation are related by an explicit specialization, without changing either series.

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